US012405450B2

(12) **United States Patent**
Liao et al.

(10) **Patent No.: US 12,405,450 B2**
(45) **Date of Patent: Sep. 2, 2025**

(54) **OPTICAL IMAGING LENS**

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(73) Assignee: **Genius Electronic Optical (Xiamen) Co., Ltd.**, Fujian (CN)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 674 days.

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(22) Filed: **Jun. 9, 2022**

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(30) **Foreign Application Priority Data**

Dec. 28, 2021 (CN) 202111626412.2

(51) **Int. Cl.**
G02B 13/00 (2006.01)
G02B 9/64 (2006.01)

(52) **U.S. Cl.**
CPC ***G02B 13/0045*** (2013.01); ***G02B 9/64*** (2013.01)

(58) **Field of Classification Search**
CPC G02B 13/0045; G02B 9/64; G02B 1/041; G02B 13/18; G03B 30/00
See application file for complete search history.

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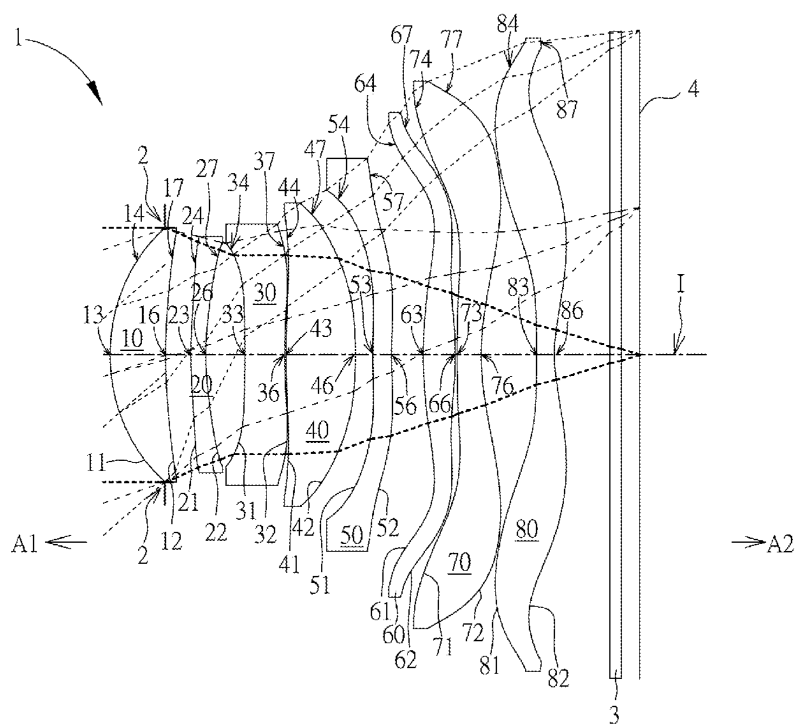
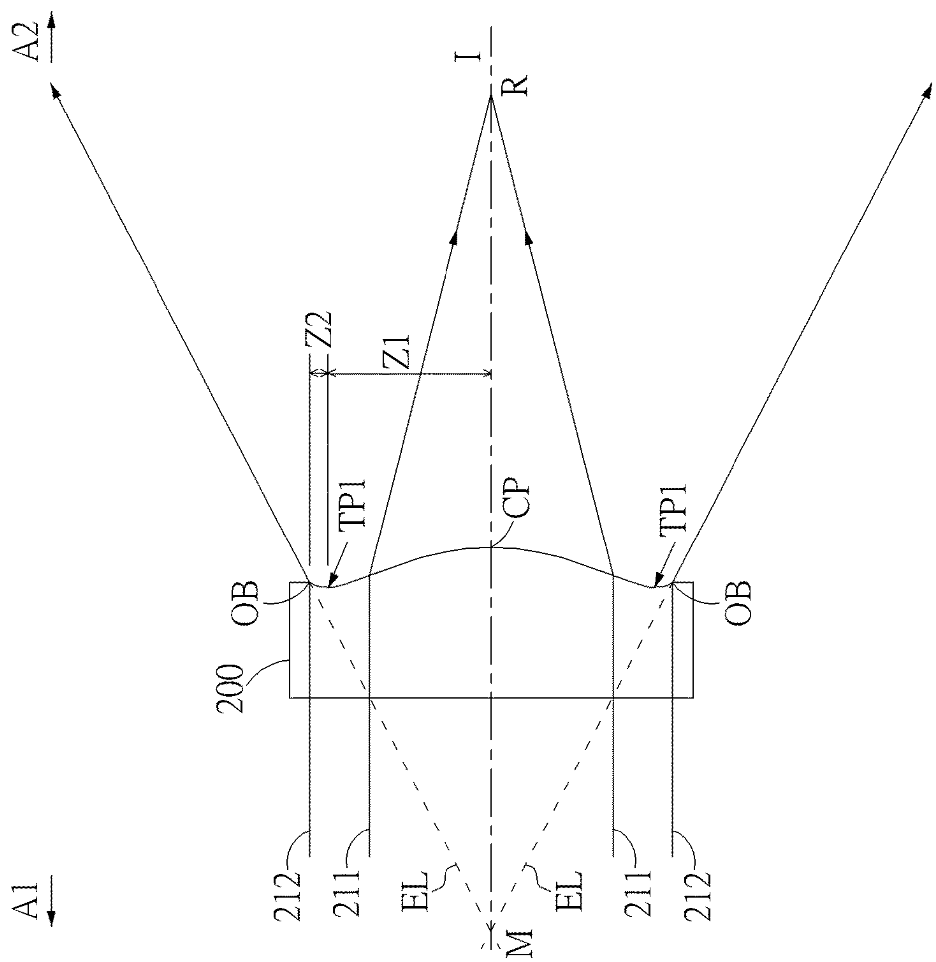
Primary Examiner — Charles S Chang
(74) *Attorney, Agent, or Firm* — Winston Hsu

(57) **ABSTRACT**

An optical imaging lens includes a first lens element to an eighth lens element and each lens element has an object-side surface and an image-side surface. The second lens element has negative refracting power, the third lens element has negative refracting power and a periphery region of the object-side surface of the third lens element is concave, the seventh lens element has negative refracting power, an optical axis region of the object-side surface of the eighth lens element is convex and a periphery region of the image-side surface of the eighth lens element is convex. Lens elements included by the optical imaging lens are only eight lens elements described above to satisfy Tavg567/Tstd567≤3.600.

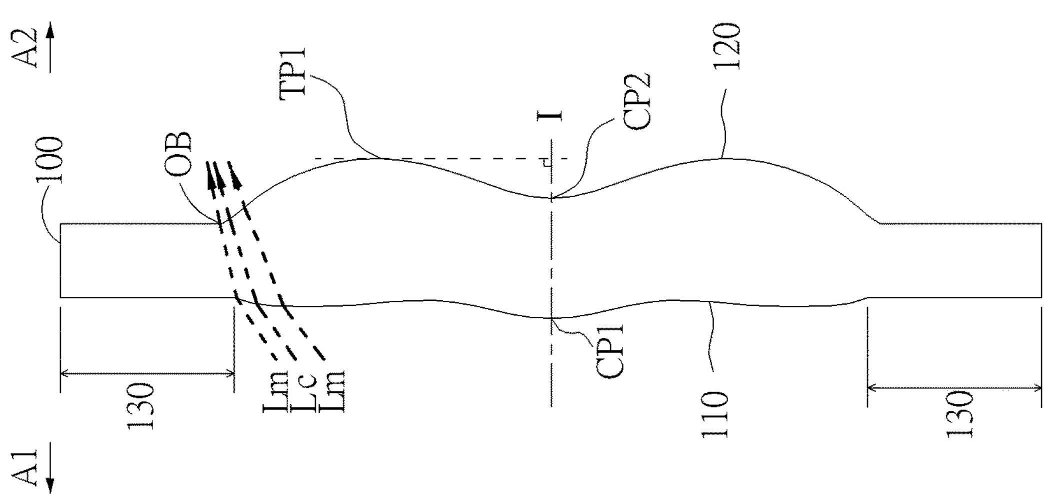
19 Claims, 25 Drawing Sheets





A2
A1
I
R
Z2
Z1
TP1
CP
TP1
OB
OB
200
212
211
EL
M
EL
211
212

FIG. 2



A2
A1
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OB
TP1
I
CP2
120
130
Lm
Lc
Lm
CP1
110
130

FIG. 1

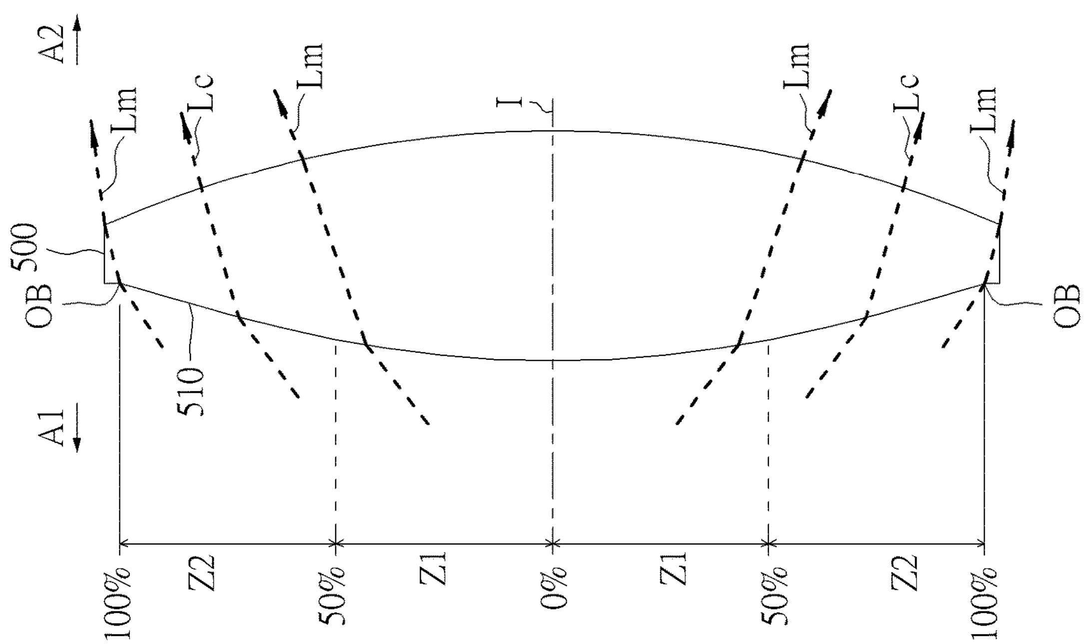

FIG. 5

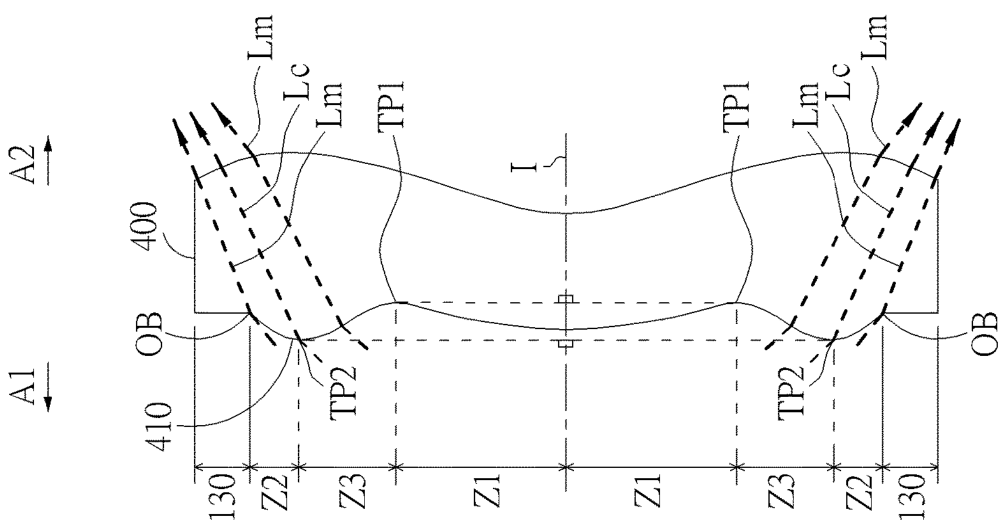

FIG. 4

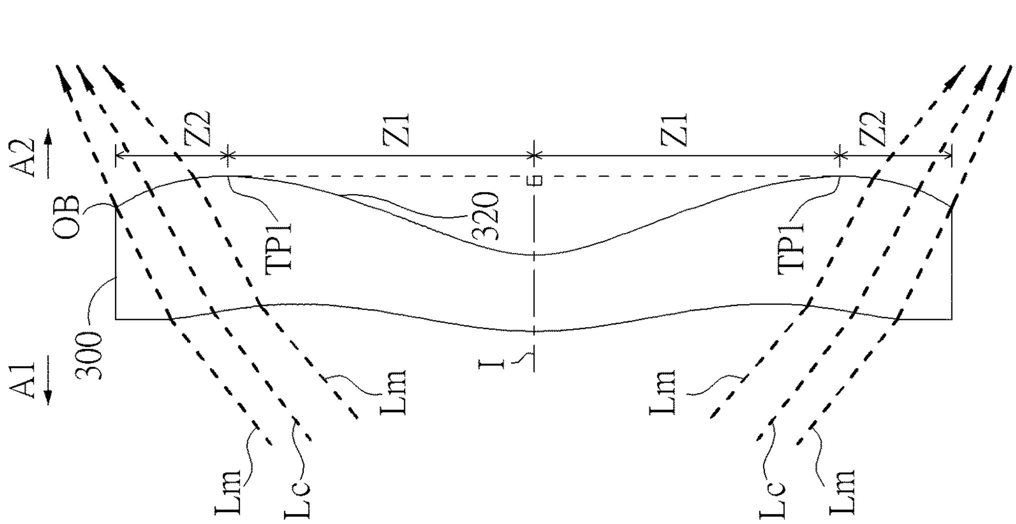

FIG. 3

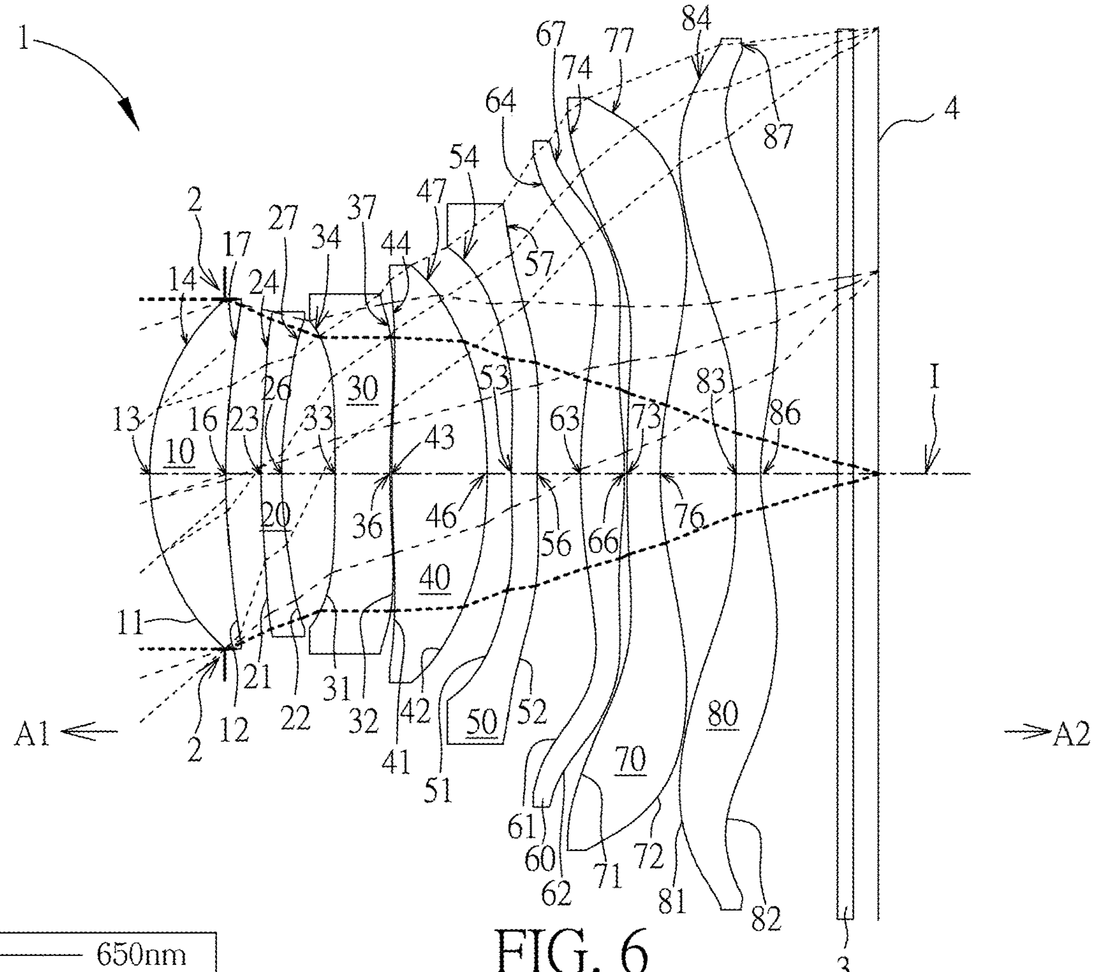

FIG. 6

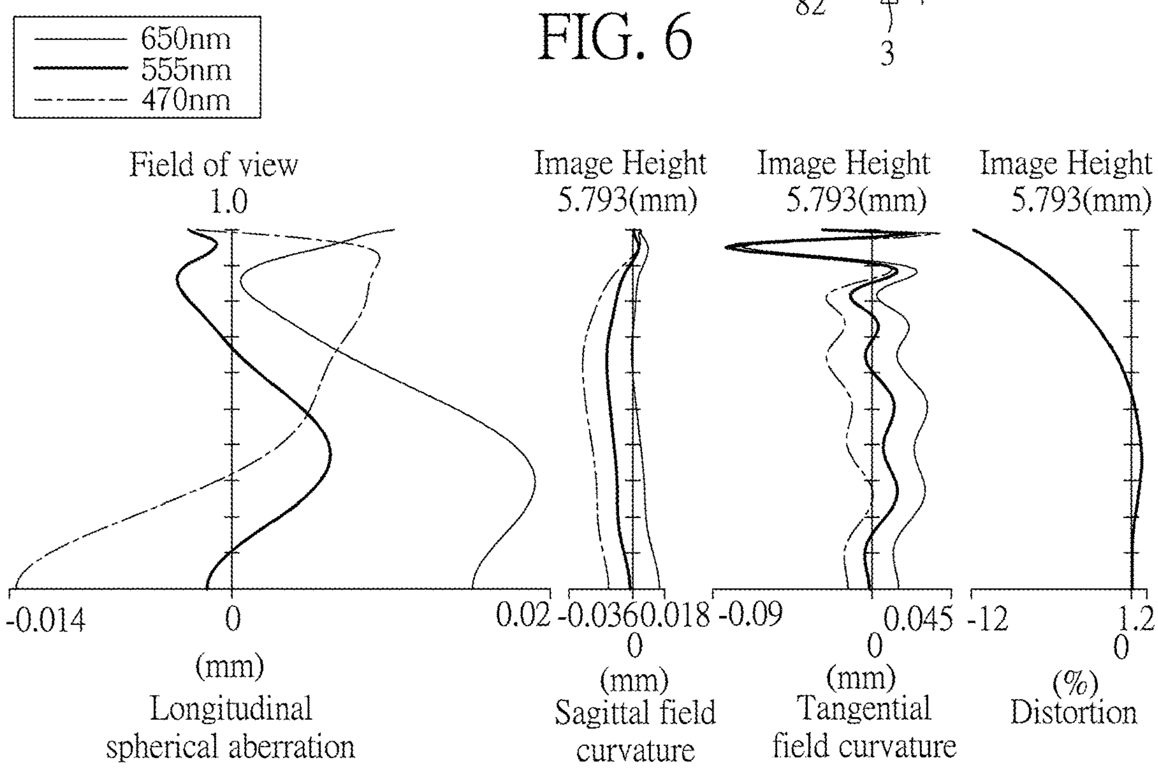

FIG. 7A FIG. 7B FIG. 7C FIG. 7D

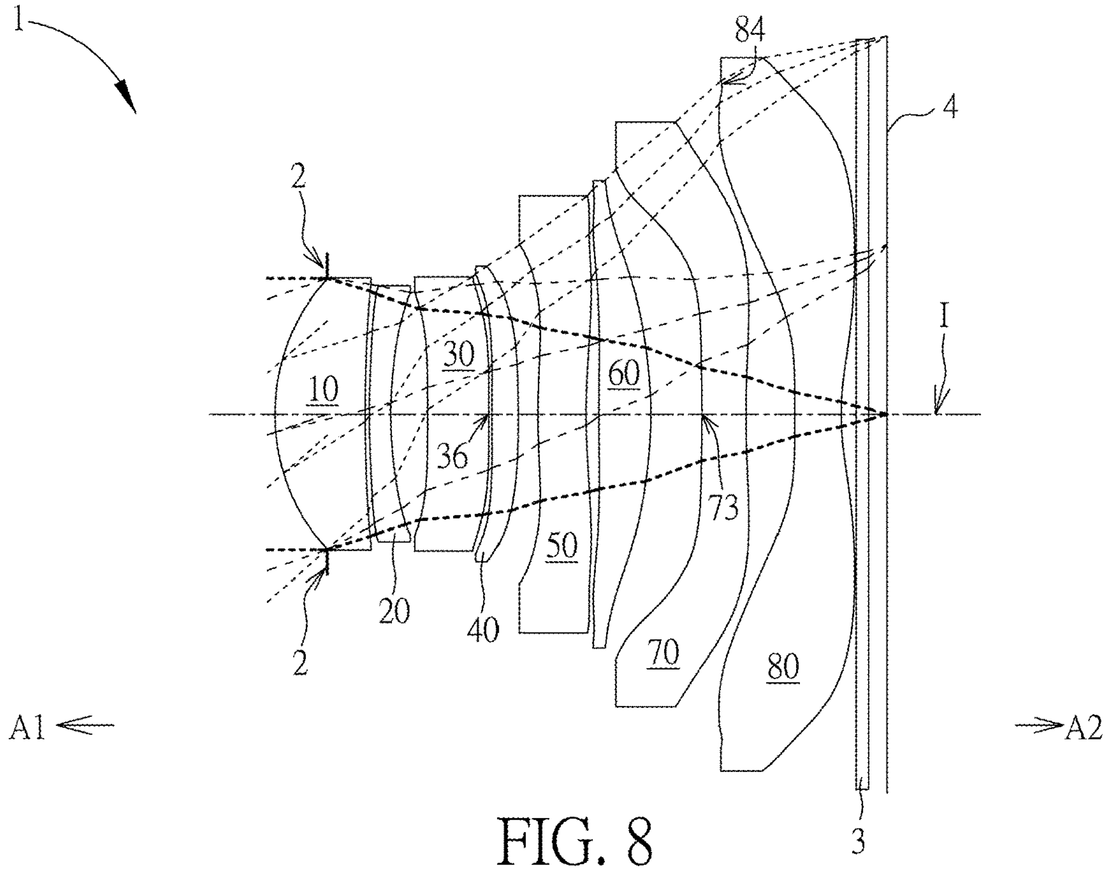

FIG. 8

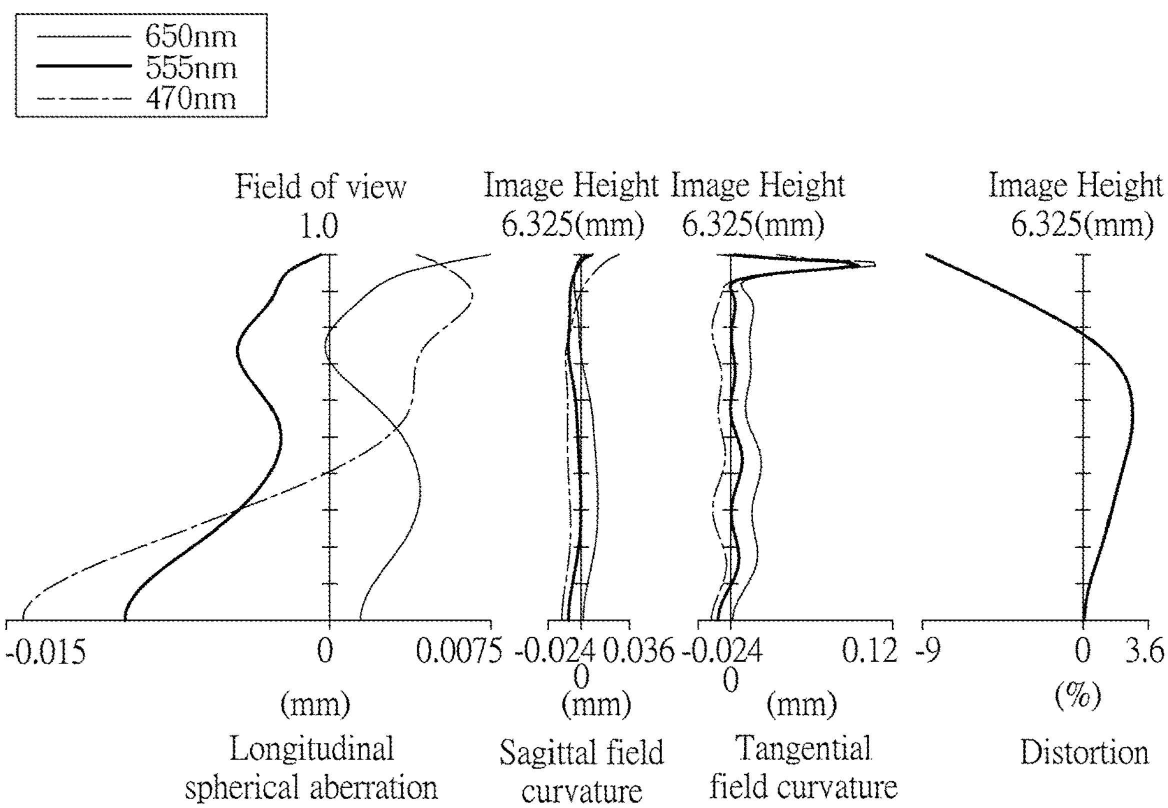

FIG. 9A FIG. 9B FIG. 9C FIG. 9D

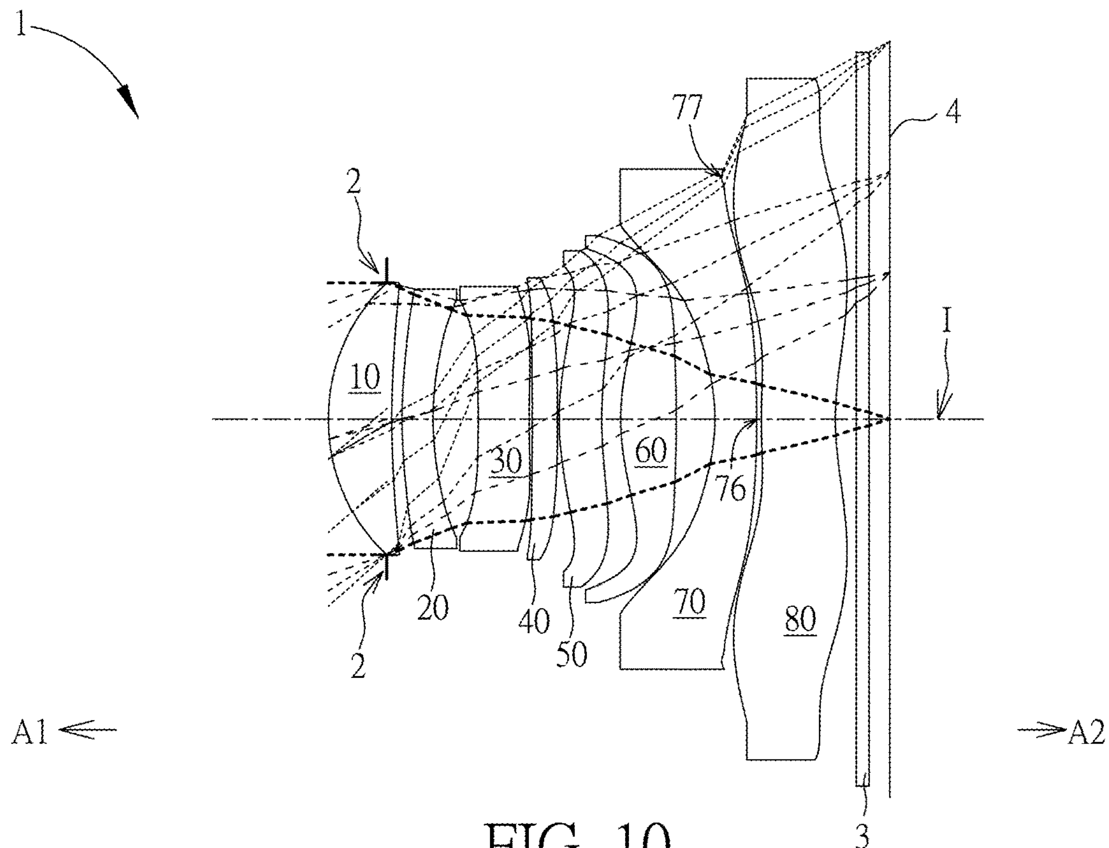

FIG. 10

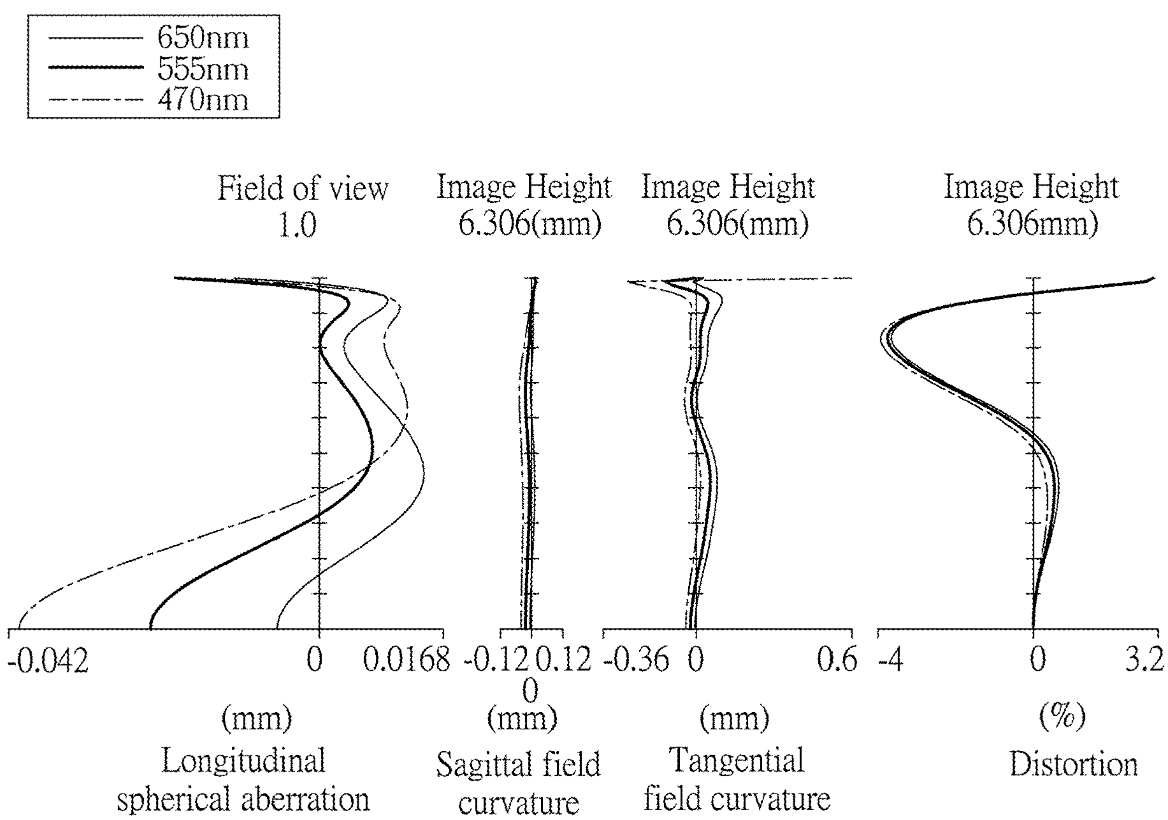

FIG. 11A FIG. 11B FIG. 11C FIG. 11D

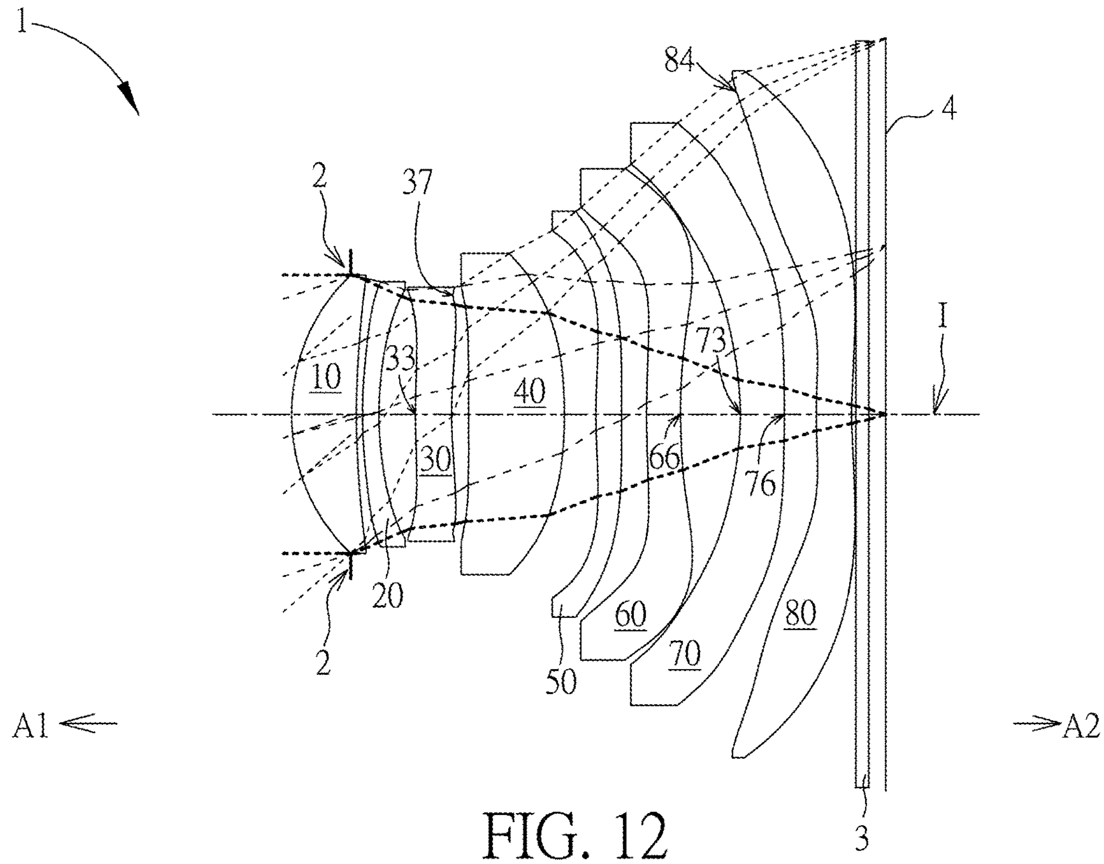

FIG. 12

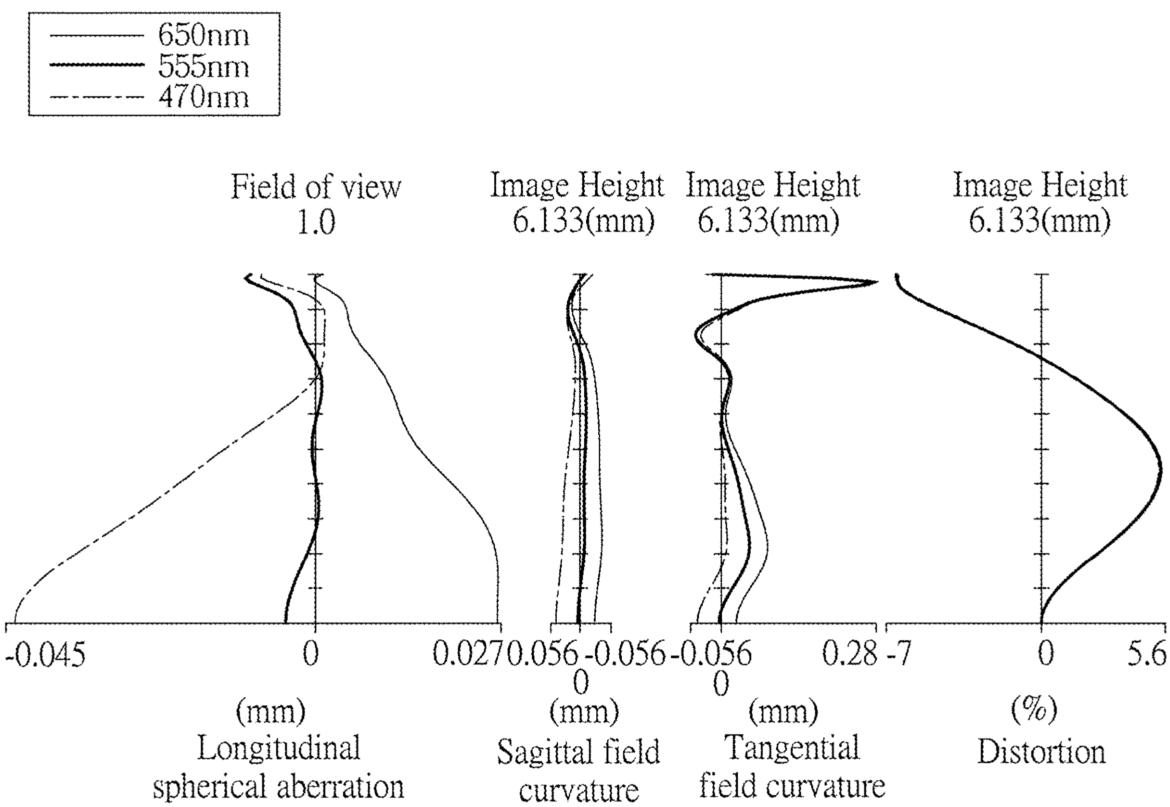

FIG. 13A FIG. 13B FIG. 13C FIG. 13D

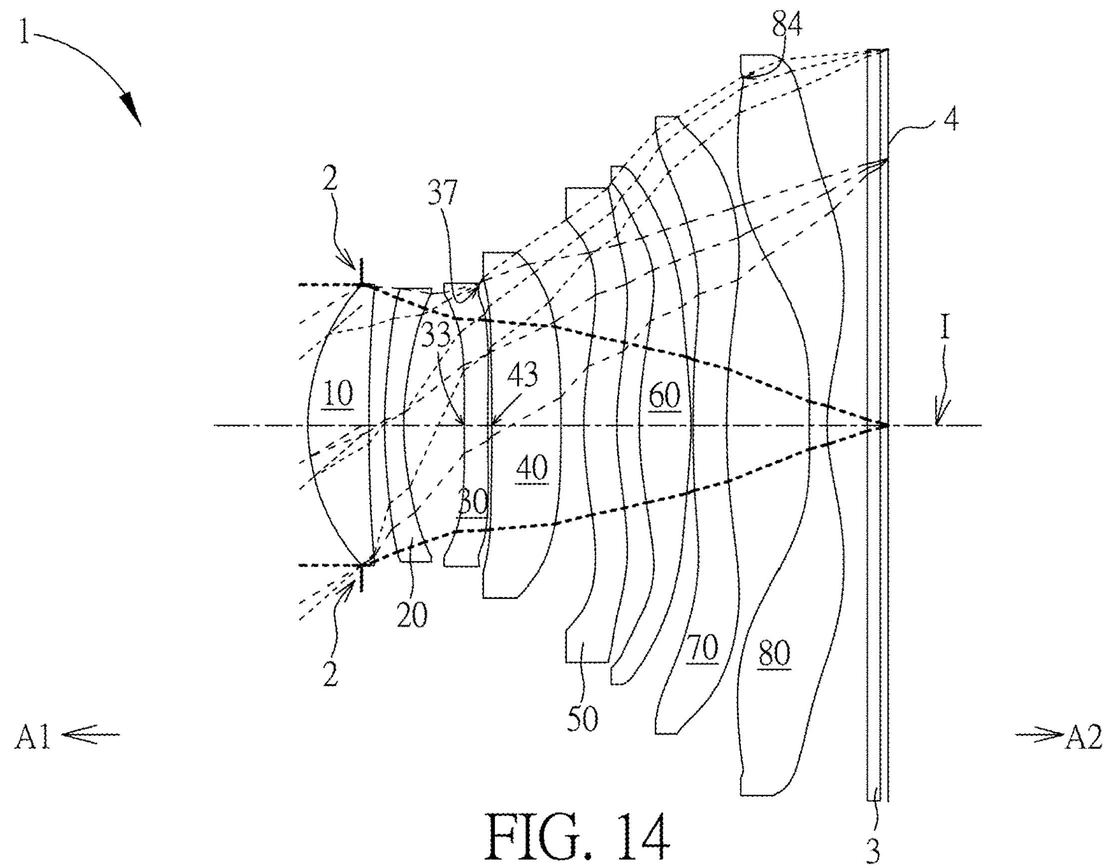

FIG. 14

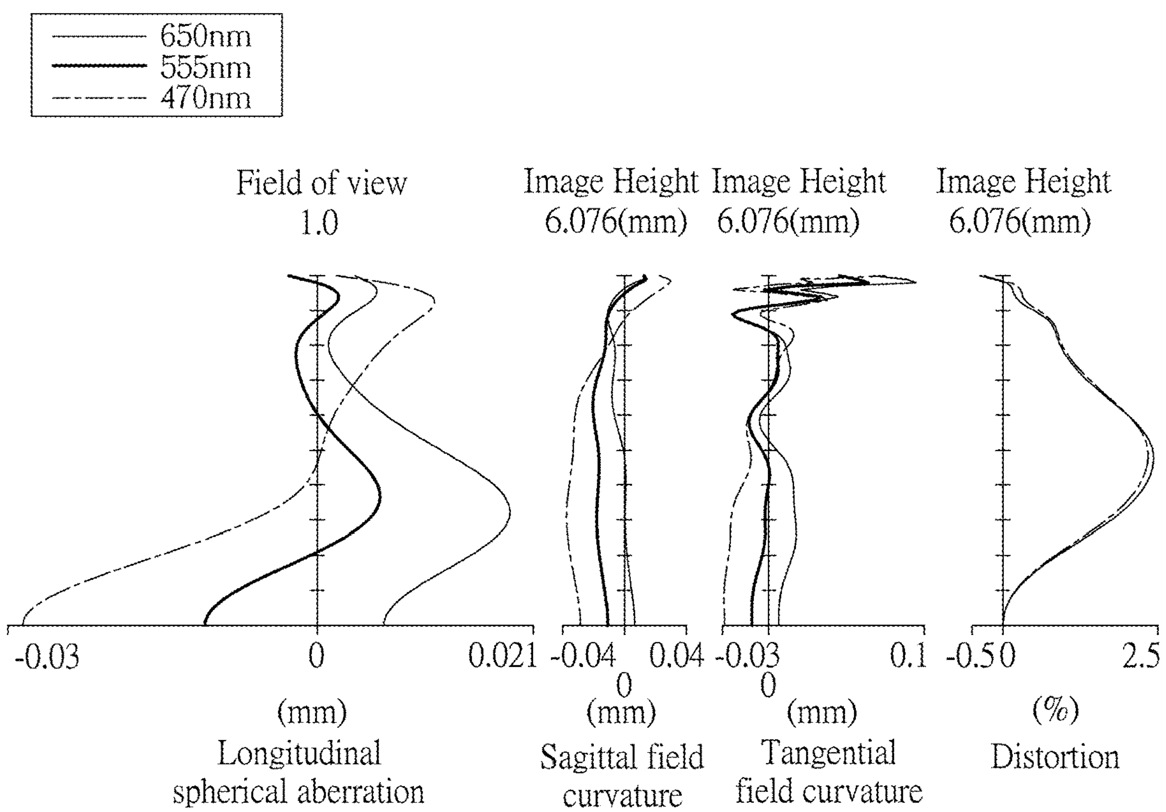

FIG. 15A FIG. 15B FIG. 15C FIG. 15D

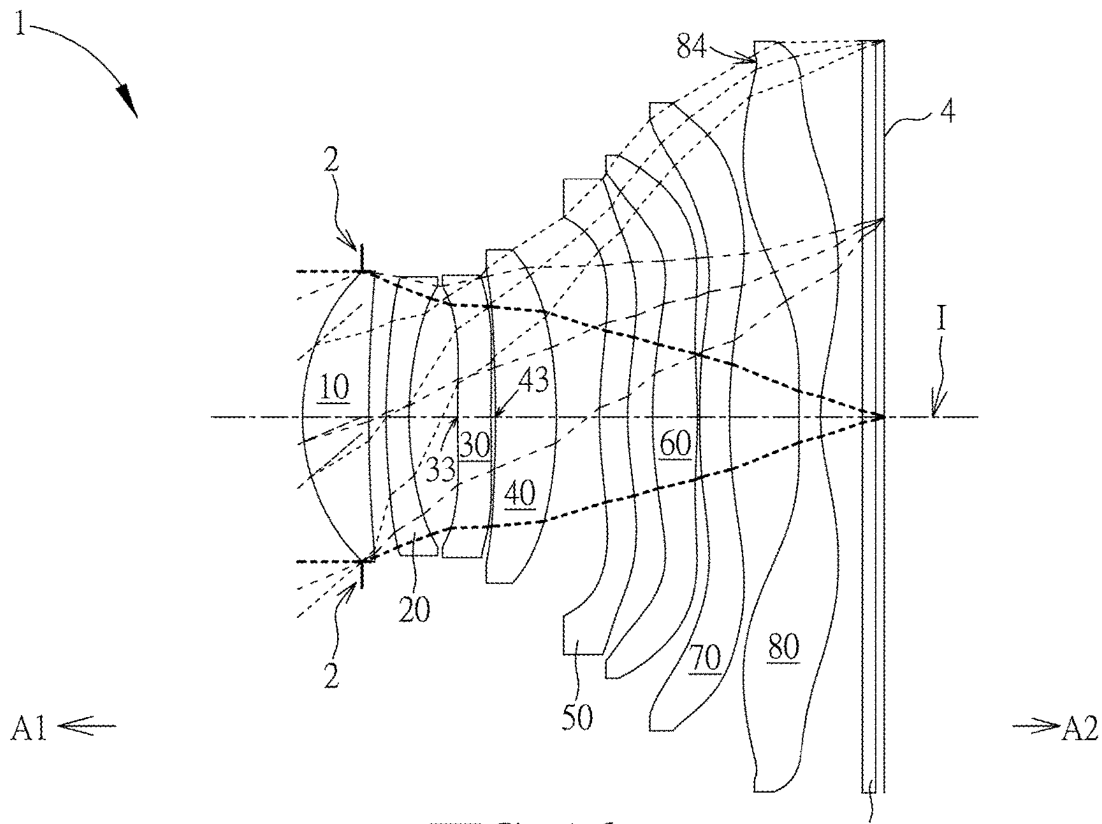

FIG. 16

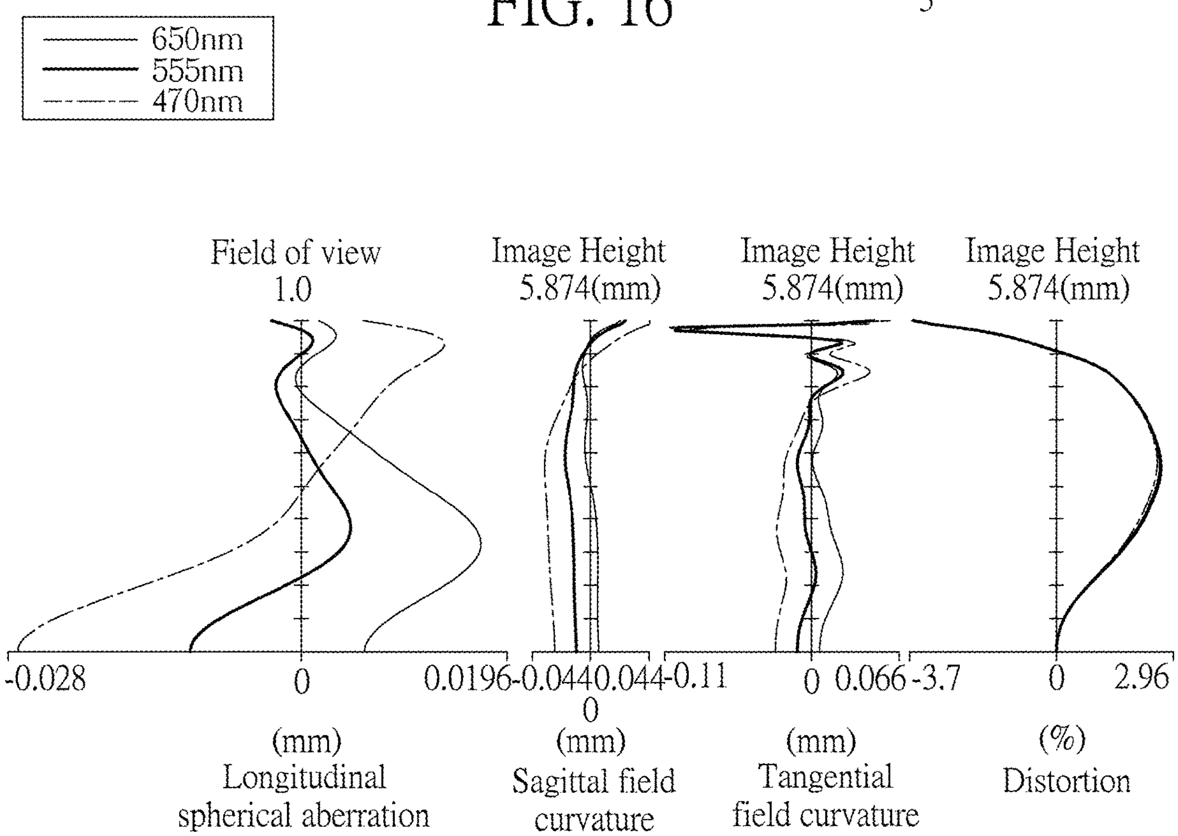

FIG. 17A FIG. 17B FIG. 17C FIG. 17D

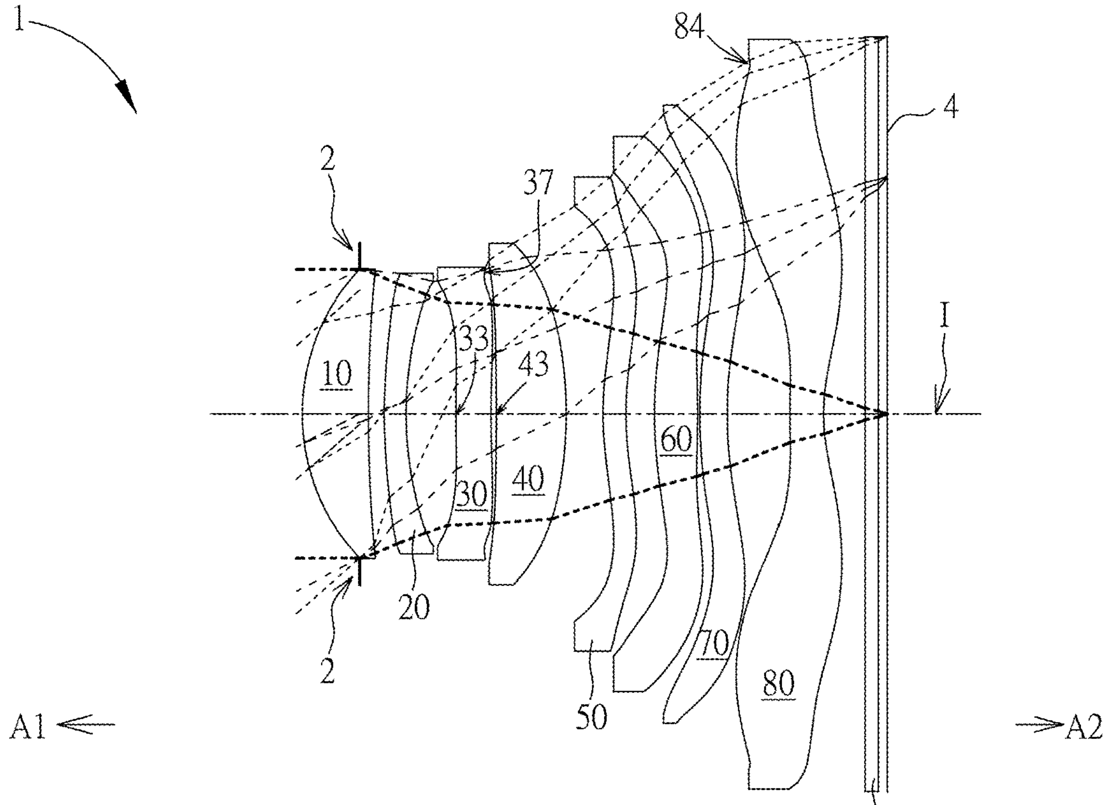

FIG. 18

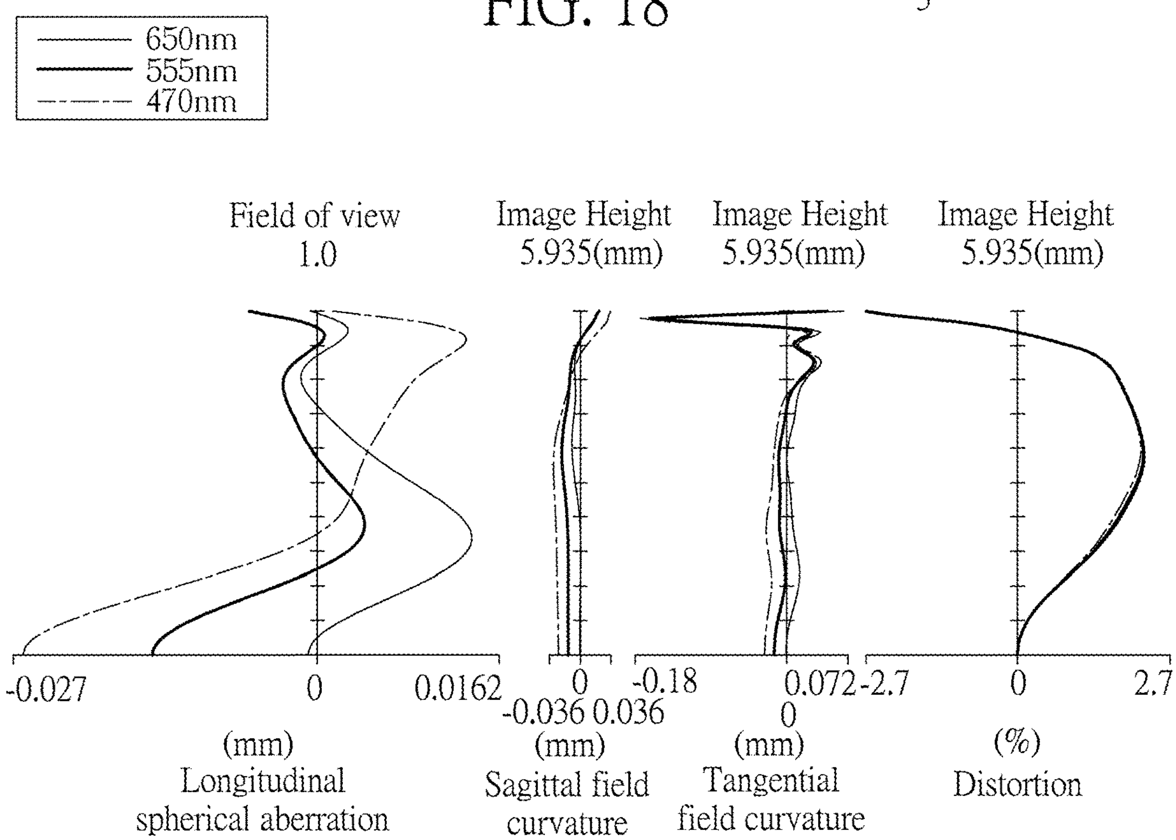

FIG. 19A FIG. 19B FIG. 19C FIG. 19D

First Embodiment							
EFL=7.598mm, HFOV=40.842 Degrees, TTL=9.477mm, Fno=1.677, ImgH=5.793mm							
No.		Radius of Curvature (mm)	Aperture Stop Distance/ Thickness/ Air Gap (mm)		Refractive Index	Abbe No.	Focal Length (mm)
	Object	Infinity	Infinity				
2	Ape. Stop	Infinity	-0.982				
11	First Lens element	3.209	0.993	T1	1.545	55.987	8.127
12		10.313	0.450	G12			
21	Second Lens element	7.330	0.270	T2	1.661	20.373	-26.408
22		5.100	0.698	G23			
31	Third Lens element	-130.479	0.713	T3	1.671	19.480	-13.310
32		9.709	0.024	G34			
41	Fourth Lens element	12.880	1.243	T4	1.567	37.533	8.683
42		-7.752	0.310	G45			
51	Fifth Lens element	12.358	0.336	T5	1.671	19.480	-185.511
52		11.128	0.566	G56			
61	Sixth Lens element	7.611	0.573	T6	1.531	49.620	6.076
62		-5.494	0.033	G67			
71	Seventh Lens element	35.973	0.434	T7	1.531	49.620	-11.004
72		5.024	0.984	G78			
81	Eighth Lens element	105.521	0.326	T8	1.531	49.620	-6.172
82		3.188	0.989	G8F			
3	Filter	Infinity	0.210		1.517	64.167	
		Infinity	0.324				
4	Image Plane	Infinity					

FIG. 20

No.	K	a_4	a_6	a_8
11	-1.270986E-01	8.483586E-04	2.494901E-04	-1.578457E-05
12	-3.114583E+01	-1.469828E-03	1.056070E-03	-4.046416E-04
21	-2.422136E+01	-1.886323E-02	5.132769E-03	-5.406430E-04
22	4.615564E+00	-2.815418E-02	5.870282E-03	-8.242850E-04
31	0.000000E+00	-1.388911E-02	-6.399040E-04	-1.864634E-04
32	0.000000E+00	-1.530905E-02	-9.360599E-04	7.681875E-04
41	0.000000E+00	-1.100038E-02	-5.061709E-04	6.912765E-04
42	0.000000E+00	-2.124443E-02	4.801855E-03	-1.294024E-03
51	-9.900741E+01	-3.506834E-02	7.632270E-03	-1.448395E-03
52	-8.972714E+00	-3.628743E-02	6.681545E-03	-7.838819E-04
61	1.650892E+00	-4.795754E-03	4.690607E-04	-1.711881E-04
62	0.000000E+00	3.895231E-02	-6.342578E-03	4.516185E-04
71	0.000000E+00	1.182235E-02	-5.045068E-03	5.924628E-04
72	0.000000E+00	-6.764639E-03	-1.107534E-03	2.240960E-04
81	0.000000E+00	-3.422994E-02	6.142648E-03	-5.546949E-04
82	-9.667574E+00	-2.072928E-02	3.066604E-03	-3.030219E-04
No.	a_{10}	a_{12}	a_{14}	a_{16}
11	1.920172E-05	-6.060912E-06	1.285393E-06	-1.041840E-07
12	1.187299E-04	-2.059730E-05	1.636331E-06	-3.449381E-08
21	1.783596E-05	5.985437E-06	-2.177054E-06	3.953281E-07
22	-9.239781E-06	1.702772E-05	-4.310978E-06	3.325367E-07
31	-1.060579E-04	5.867761E-05	-1.682493E-05	1.982118E-06
32	-2.674173E-04	5.368166E-05	-7.091041E-06	5.244178E-07
41	-8.022790E-05	-1.074912E-05	2.205739E-06	-8.042936E-08
42	2.906999E-04	-3.144686E-05	6.459744E-07	5.653507E-08
51	2.458433E-04	-2.940004E-05	1.745959E-06	-3.645755E-08
52	6.968438E-05	-3.904220E-06	9.914487E-08	-4.443840E-10
61	-5.914446E-07	1.815585E-06	-1.101636E-07	1.930746E-09
62	-1.469356E-05	-1.281883E-07	2.931241E-08	-7.091710E-10
71	-3.328506E-05	9.429736E-07	-1.136478E-08	2.007800E-11
72	-1.982516E-05	9.446586E-07	-2.400396E-08	2.488010E-10
81	2.933984E-05	-9.118740E-07	1.548870E-08	-1.113930E-10
82	1.790738E-05	-5.963646E-07	1.049919E-08	-7.698300E-11

FIG. 21

Second Embodiment							
EFL=8.000mm, HFOV=40.842 Degrees, TTL=10.270mm, Fno=1.766, ImgH=6.325mm							
No.		Radius of Curvature (mm)	Aperture Stop Distance/ Thickness/ Air Gap (mm)		Refractive Index	Abbe No.	Focal Length (mm)
	Object	Infinity	Infinity				
2	Ape. Stop	Infinity	-0.875				
11	First Lens element	3.428	1.514	T1	1.545	55.987	7.290
12		20.771	0.073	G12			
21	Second Lens element	13.799	0.358	T2	1.661	20.373	-19.189
22		6.572	0.625	G23			
31	Third Lens element	-48.162	1.027	T3	1.661	20.373	-100.926
32		-170.614	0.054	G34			
41	Fourth Lens element	72.502	0.431	T4	1.545	55.987	33.703
42		-24.626	0.330	G45			
51	Fifth Lens element	10.703	0.807	T5	1.661	20.373	-83.955
52		8.715	0.218	G56			
61	Sixth Lens element	45.827	0.874	T6	1.545	55.987	10.360
62		-6.413	0.858	G67			
71	Seventh Lens element	-34.723	0.742	T7	1.661	20.373	-18.487
72		19.291	0.809	G78			
81	Eighth Lens element	67.264	0.778	T8	1.545	55.987	-8.082
82		4.126	0.253	G8F			
3	Filter	Infinity	0.210		1.517	64.167	
		Infinity	0.309				
4	Image Plane	Infinity					

FIG. 22

No.	K	a_4	a_6	a_8
11	-1.308865E-01	9.136043E-04	1.607270E-04	-4.763881E-05
12	2.009730E+01	-6.328718E-03	2.774172E-03	-5.648340E-04
21	6.193622E+00	-1.134901E-02	4.189429E-03	-7.390547E-04
22	7.396944E+00	-8.901422E-03	1.681684E-03	-2.980330E-04
31	0.000000E+00	-7.849437E-03	-7.759088E-04	1.521547E-04
32	0.000000E+00	-1.436726E-02	1.505758E-03	-4.014160E-04
41	0.000000E+00	-1.497805E-02	3.071649E-03	-6.048335E-04
42	0.000000E+00	-1.738519E-02	5.165551E-03	-1.406514E-03
51	-4.560972E+01	-1.733993E-02	5.439586E-03	-1.541601E-03
52	3.023828E-01	-1.973023E-02	5.684146E-03	-1.179892E-03
61	9.899930E+01	-6.151303E-03	8.619178E-04	-1.121503E-04
62	0.000000E+00	1.277821E-02	-3.698075E-03	4.559409E-04
71	0.000000E+00	7.363123E-03	-3.521730E-03	4.313761E-04
72	0.000000E+00	-1.162570E-03	-1.203124E-03	2.034611E-04
81	0.000000E+00	-3.690605E-02	6.187225E-03	-5.527525E-04
82	-5.920655E+00	-2.091486E-02	3.367517E-03	-3.161885E-04
No.	a_{10}	a_{12}	a_{14}	a_{16}
11	3.078300E-05	-8.394380E-06	1.222122E-06	-7.382612E-08
12	2.682901E-05	9.945087E-06	-1.412039E-06	3.415294E-08
21	2.739224E-05	1.946903E-05	-3.157198E-06	1.724676E-07
22	-2.896498E-05	2.567860E-05	-5.422508E-06	4.407567E-07
31	-1.285273E-04	5.058635E-05	-9.436052E-06	7.509742E-07
32	5.183773E-05	9.619982E-06	-3.712185E-06	3.822261E-07
41	1.504154E-05	1.481489E-05	-4.760665E-06	4.994114E-07
42	2.198110E-04	-2.603877E-05	1.653532E-06	-5.721441E-09
51	2.255404E-04	-1.815840E-05	1.031888E-06	-4.262771E-08
52	1.340379E-04	-7.569023E-06	1.797903E-07	-8.945620E-10
61	4.269988E-06	1.008623E-06	-1.067267E-07	2.863327E-09
62	-2.152769E-05	-1.521308E-07	4.982926E-08	-1.316323E-09
71	-3.287649E-05	1.299810E-06	-4.525817E-09	-7.042470E-10
72	-1.857751E-05	9.411381E-07	-2.423363E-08	2.469960E-10
81	2.931636E-05	-9.137055E-07	1.547495E-08	-1.101940E-10
82	1.780471E-05	-5.933316E-07	1.064707E-08	-7.858000E-11

FIG. 23

Third Embodiment							
EFL=7.038mm, HFOV=40.842 Degrees, TTL=9.398mm, Fno=1.554, ImgH=6.306mm							
No.		Radius of Curvature (mm)	Aperture Stop Distance/ Thickness/ Air Gap (mm)		Refractive Index	Abbe No.	Focal Length (mm)
	Object	Infinity	Infinity				
2	Ape. Stop	Infinity	-0.982				
11	First Lens element	3.168	1.065	T1	1.545	55.987	7.575
12		11.911	0.173	G12			
21	Second Lens element	7.299	0.522	T2	1.661	20.373	-26.983
22		5.045	0.755	G23			
31	Third Lens element	-36.436	0.847	T3	1.661	20.373	-18.157
32		18.314	0.051	G34			
41	Fourth Lens element	-62.102	0.424	T4	1.545	55.987	4011.516
42		-60.536	0.025	G45			
51	Fifth Lens element	4.628	0.715	T5	1.661	20.373	20.867
52		6.503	0.322	G56			
61	Sixth Lens element	3.822	0.934	T6	1.545	55.987	5.628
62		-14.368	0.644	G67			
71	Seventh Lens element	-3.177	0.704	T7	1.661	20.373	-6.201
72		-14.913	0.077	G78			
81	Eighth Lens element	10.058	1.229	T8	1.545	55.987	-16.740
82		4.583	0.355	G8F			
3	Filter	Infinity	0.210		1.517	64.167	
		Infinity	0.346				
4	Image Plane	Infinity					

FIG. 24

No.	K	a_4	a_6	a_8
11	-4.799018E-02	1.173520E-04	5.307787E-04	-1.990610E-04
12	-1.104944E+02	-4.996992E-03	1.986111E-03	-2.077795E-04
21	-1.989425E+01	-1.358585E-02	5.182532E-03	-7.382885E-04
22	4.463857E+00	-1.425351E-02	3.171337E-03	-5.311220E-04
31	3.570870E+00	-1.420590E-02	-1.324416E-03	6.921709E-04
32	-2.000050E+01	-1.515964E-02	-1.458615E-03	7.676099E-04
41	-1.997975E+01	2.496388E-03	-1.127074E-03	3.547555E-04
42	-2.007615E+01	-1.165251E-02	3.058209E-03	-7.826747E-04
51	-2.332918E+00	-1.616062E-02	5.234006E-03	-1.541380E-03
52	2.100431E+00	-2.375982E-02	6.128437E-03	-1.426484E-03
61	-5.166593E+00	-9.279734E-03	8.913267E-04	-2.398323E-04
62	1.905104E+01	1.432174E-02	-3.527025E-03	2.600125E-04
71	-4.712194E+00	1.113045E-02	-5.568939E-03	6.102456E-04
72	9.133558E+00	3.962623E-03	-1.771373E-03	2.606183E-04
81	1.069633E+00	-4.098905E-02	6.652810E-03	-5.656895E-04
82	-1.548984E+00	-2.362923E-02	3.219866E-03	-3.070101E-04
No.	a_{10}	a_{12}	a_{14}	a_{16}
11	5.481635E-05	-5.851459E-06	2.398211E-07	-2.788023E-09
12	-1.548821E-05	3.931882E-06	-1.218777E-09	-2.130445E-08
21	6.683394E-07	1.076037E-05	-3.580364E-07	-2.405054E-08
22	-4.044701E-05	1.195395E-05	-3.438668E-07	-1.301328E-07
31	-3.026645E-04	4.647929E-05	-1.675518E-08	-1.176871E-07
32	-2.669863E-04	4.568737E-05	-1.291049E-06	-6.603668E-08
41	-9.417836E-05	8.240586E-06	-4.447045E-07	6.311935E-08
42	2.061057E-04	-3.828865E-05	1.595385E-06	9.771423E-08
51	2.172147E-04	-1.650364E-05	8.073423E-07	-7.464834E-08
52	1.431118E-04	-6.408007E-06	1.461107E-07	-1.394698E-08
61	-2.352999E-06	1.924000E-07	-1.534883E-07	1.068762E-08
62	-1.618664E-05	1.432315E-07	3.068423E-09	-2.286178E-09
71	-3.079967E-05	1.099631E-06	-1.227065E-08	-8.149480E-10
72	-1.940796E-05	8.888781E-07	-2.508819E-08	3.446230E-10
81	2.903842E-05	-9.062719E-07	1.590817E-08	-1.212110E-10
82	1.794159E-05	-5.986383E-07	1.054833E-08	-7.671800E-11

FIG. 25

Fourth Embodiment							
EFL=7.587mm, HFOV=40.842 Degrees, TTL=9.721mm, Fno=1.675, ImgH=6.133mm							
No.		Radius of Curvature (mm)	Aperture Stop Distance/ Thickness/ Air Gap (mm)		Refractive Index	Abbe No.	Focal Length (mm)
	Object	Infinity	Infinity				
2	Ape. Stop	Infinity	-0.966				
11	First Lens element	3.210	1.057	T1	1.545	55.987	8.168
12		10.107	0.103	G12			
21	Second Lens element	7.178	0.267	T2	1.661	20.373	-32.206
22		5.300	0.599	G23			
31	Third Lens element	13.316	0.594	T3	1.661	20.373	-37.868
32		8.564	0.273	G34			
41	Fourth Lens element	205.607	1.571	T4	1.545	55.987	12.981
42		-7.325	0.513	G45			
51	Fifth Lens element	9.909	0.419	T5	1.661	20.373	18.624
52		48.124	0.410	G56			
61	Sixth Lens element	16.605	0.560	T6	1.545	55.987	-200.000
62		14.243	0.976	G67			
71	Seventh Lens element	-4.501	0.721	T7	1.661	20.373	-8.026
72		-30.039	0.519	G78			
81	Eighth Lens element	8.279	0.578	T8	1.545	55.987	199.988
82		8.737	0.071	G8F			
3	Filter	Infinity	0.210		1.517	64.167	
		Infinity	0.279				
4	Image Plane	Infinity					

FIG. 26

No.	K	a_4	a_6	a_8
11	-1.269481E-02	2.954823E-04	5.225206E-04	-2.177074E-04
12	-7.544467E+01	-3.832825E-03	1.629804E-03	-1.573693E-04
21	-2.305986E+01	-1.086611E-02	5.547808E-03	-7.801332E-04
22	4.646672E+00	-1.330024E-02	3.981298E-03	-5.666561E-04
31	0.000000E+00	-1.600104E-02	-1.201226E-03	8.050258E-04
32	0.000000E+00	-1.229062E-02	-1.906294E-03	9.074882E-04
41	0.000000E+00	-2.106585E-03	-1.715795E-03	2.013376E-04
42	0.000000E+00	-1.378191E-02	2.866569E-03	-9.973650E-04
51	-1.329780E+01	-2.318867E-02	5.766075E-03	-1.450552E-03
52	9.899999E+01	-2.102128E-02	6.146044E-03	-1.292381E-03
61	1.782936E+01	-6.919444E-03	3.919416E-04	-2.232046E-04
62	0.000000E+00	1.285412E-02	-3.973291E-03	3.751045E-04
71	0.000000E+00	1.956613E-02	-4.933597E-03	5.867341E-04
72	0.000000E+00	5.295919E-04	-1.853968E-03	2.681734E-04
81	0.000000E+00	-4.621498E-02	6.877098E-03	-5.672138E-04
82	7.741415E-01	-1.981465E-02	3.030932E-03	-3.052304E-04
No.	a_{10}	a_{12}	a_{14}	a_{16}
11	5.698909E-05	-5.217414E-06	2.090151E-08	6.479208E-09
12	-2.126606E-05	4.316738E-06	2.076580E-08	-2.152188E-08
21	-5.960832E-06	1.102029E-05	-1.950747E-08	2.700602E-09
22	-5.168090E-05	1.049178E-05	3.338854E-07	-4.898975E-08
31	-3.106620E-04	4.969849E-05	-4.364042E-07	-2.880472E-07
32	-2.784536E-04	4.697123E-05	-5.331550E-07	-9.622979E-08
41	-7.959973E-05	1.444447E-05	2.323975E-07	1.802515E-08
42	2.362038E-04	-3.222185E-05	1.930514E-06	-1.951987E-08
51	2.153157E-04	-1.763251E-05	5.546946E-07	2.344994E-09
52	1.397193E-04	-7.310080E-06	1.464386E-07	-4.952900E-11
61	4.210459E-06	1.583969E-06	-7.196682E-08	1.886600E-10
62	-1.510773E-05	-1.994853E-07	3.378333E-08	-6.913060E-10
71	-3.396052E-05	9.384759E-07	-1.089210E-08	6.356800E-11
72	-1.940829E-05	8.834546E-07	-2.512626E-08	3.256190E-10
81	2.896244E-05	-9.086897E-07	1.591228E-08	-1.180260E-10
82	1.790619E-05	-6.000674E-07	1.062746E-08	-7.723200E-11

FIG. 27

Fifth Embodiment							
EFL=7.036mm, HFOV=40.842 Degrees, TTL=9.407mm, Fno=1.553, ImgH=6.076mm							
No.		Radius of Curvature (mm)	Aperture Stop Distance/ Thickness/ Air Gap (mm)		Refractive Index	Abbe No.	Focal Length (mm)
	Object	Infinity	Infinity				
2	Ape. Stop	Infinity	-0.869				
11	First Lens element	3.454	0.991	T1	1.545	55.987	7.176
12		26.111	0.252	G12			
21	Second Lens element	7.368	0.315	T2	1.661	20.373	-20.091
22		4.673	0.982	G23			
31	Third Lens element	154.448	0.384	T3	1.661	20.373	-100.156
32		46.593	0.067	G34			
41	Fourth Lens element	-28.605	1.108	T4	1.545	55.987	-431.209
42		-33.004	0.371	G45			
51	Fifth Lens element	5.282	0.539	T5	1.661	20.373	-132.066
52		4.780	0.367	G56			
61	Sixth Lens element	6.083	0.839	T6	1.545	55.987	5.860
62		-6.427	0.035	G67			
71	Seventh Lens element	64.163	0.547	T7	1.661	20.373	-33.516
72		16.525	1.331	G78			
81	Eighth Lens element	33.720	0.287	T8	1.545	55.987	-6.002
82		2.979	0.655	G8F			
3	Filter	Infinity	0.210		1.517	64.167	
		Infinity	0.129				
4	Image Plane	Infinity					

FIG. 28

No.	K	a_4	a_6	a_8
11	-7.670019E-02	6.857721E-04	3.414066E-04	-1.379192E-04
12	2.530735E+01	-3.310259E-03	1.258652E-03	-1.896084E-04
21	-9.400392E+00	-1.386923E-02	4.873116E-03	-7.102592E-04
22	3.009662E+00	-1.831669E-02	3.775838E-03	-5.185829E-04
31	9.947554E-01	-1.426161E-02	-2.187590E-03	9.701462E-04
32	3.336119E+01	-3.828808E-03	-2.852586E-03	1.009862E-03
41	-1.999953E+01	7.341214E-03	-1.750313E-03	1.054512E-04
42	5.000002E+01	-1.029328E-02	2.025310E-03	-9.231170E-04
51	-5.816525E+00	-1.796652E-02	5.857387E-03	-1.522833E-03
52	-1.138121E+00	-2.752269E-02	6.379098E-03	-1.280256E-03
61	1.326488E+00	-9.492179E-03	1.152070E-03	-2.460673E-04
62	1.291387E+00	2.429897E-02	-4.706425E-03	4.292309E-04
71	4.023435E+01	1.633513E-02	-5.073409E-03	5.771035E-04
72	1.940683E+00	7.766782E-03	-2.319850E-03	2.640845E-04
81	3.297464E+01	-4.426958E-02	6.885001E-03	-5.662529E-04
82	-6.730713E+00	-2.085325E-02	3.112519E-03	-3.043752E-04

No.	a_{10}	a_{12}	a_{14}	a_{16}
11	4.229733E-05	-6.197478E-06	4.819648E-07	-2.720170E-08
12	2.518062E-07	4.229838E-06	-6.699767E-07	3.829542E-08
21	2.062789E-05	9.828404E-06	-1.146788E-06	8.165491E-08
22	-3.031732E-05	1.126415E-05	-6.898586E-07	-3.972515E-09
31	-2.916393E-04	4.071656E-05	-1.581185E-06	6.567270E-08
32	-2.986075E-04	4.108164E-05	-8.247183E-07	-3.561552E-08
41	-6.169978E-05	1.202304E-05	-1.264081E-06	1.205141E-07
42	2.314286E-04	-3.172999E-05	1.903310E-06	-2.859363E-08
51	2.156388E-04	-1.669752E-05	6.091206E-07	-6.560793E-09
52	1.399095E-04	-7.315284E-06	1.456075E-07	-8.550000E-13
61	4.568522E-06	1.497446E-06	-7.956455E-08	5.326170E-10
62	-1.568784E-05	-2.323462E-07	3.382255E-08	-6.637740E-10
71	-3.338597E-05	9.662684E-07	-1.068501E-08	-1.272100E-11
72	-1.923997E-05	9.063811E-07	-2.465568E-08	2.863380E-10
81	2.890099E-05	-9.095886E-07	1.603600E-08	-1.204910E-10
82	1.792496E-05	-6.009339E-07	1.058738E-08	-7.632400E-11

FIG. 29

Sixth Embodiment								
EFL=7.031mm, HFOV=40.842 Degrees, TTL=9.118mm, Fno=1.552, ImgH=5.874mm								
No.		Radius of Curvature (mm)	Aperture Stop Distance/ Thickness/ Air Gap (mm)		Refractive Index	Abbe No.	Focal Length (mm)
	Object	Infinity	Infinity				
2	Ape. Stop	Infinity	-0.927				
11	First Lens element	3.288	1.034	T1	1.545	55.987	6.929
12		22.253	0.272	G12			
21	Second Lens element	9.055	0.358	T2	1.661	20.373	-16.072
22		4.832	0.764	G23			
31	Third Lens element	27.832	0.519	T3	1.661	20.373	223.349
32		33.970	0.073	G34			
41	Fourth Lens element	-17.854	0.957	T4	1.545	55.987	29.177
42		-8.581	0.676	G45			
51	Fifth Lens element	6.116	0.441	T5	1.661	20.373	-51.294
52		5.039	0.392	G56			
61	Sixth Lens element	7.672	0.699	T6	1.545	55.987	7.597
62		-8.748	0.035	G67			
71	Seventh Lens element	17.943	0.466	T7	1.661	20.373	-34.957
72		10.035	1.085	G78			
81	Eighth Lens element	24.140	0.348	T8	1.545	55.987	-6.099
82		2.913	0.655	G8F			
3	Filter	Infinity	0.210		1.517	64.167	
		Infinity	0.134				
4	Image Plane	Infinity					

FIG. 30

No.	K	a_4	a_6	a_8
11	-4.978248E-02	4.230685E-04	4.053853E-04	-1.678176E-04
12	1.027016E+01	-3.505762E-03	1.252677E-03	-1.540514E-04
21	-1.516832E+01	-1.277818E-02	5.159405E-03	-7.643614E-04
22	3.540359E+00	-1.668235E-02	4.242265E-03	-5.927612E-04
31	5.000159E+01	-1.399479E-02	-2.578079E-03	8.888585E-04
32	4.893989E+01	-5.698546E-03	-2.835440E-03	9.822399E-04
41	-2.000061E+01	4.349094E-03	-8.159806E-04	1.262536E-04
42	-1.976980E+01	-1.034730E-02	2.019577E-03	-8.418950E-04
51	-2.285648E+00	-2.203888E-02	5.995992E-03	-1.508864E-03
52	-1.379293E+00	-2.854358E-02	6.279388E-03	-1.272877E-03
61	2.828805E+00	-4.419843E-03	4.946718E-04	-2.381304E-04
62	-1.558260E+00	2.927147E-02	-5.403634E-03	4.316052E-04
71	4.194958E+00	1.501563E-02	-5.134041E-03	5.801712E-04
72	-1.278432E+00	5.003584E-03	-2.272439E-03	2.707530E-04
81	1.643926E+01	-4.211703E-02	6.783331E-03	-5.680232E-04
82	-6.935127E+00	-2.077120E-02	3.133648E-03	-3.057177E-04
No.	a_{10}	a_{12}	a_{14}	a_{16}
11	4.945868E-05	-6.177673E-06	2.398479E-07	6.412670E-10
12	-8.759667E-06	4.367638E-06	-4.414052E-07	1.708423E-08
21	1.079344E-05	1.190654E-05	-4.488128E-07	-4.950305E-08
22	-3.544922E-05	1.323108E-05	-2.852310E-07	-2.744288E-08
31	-2.735872E-04	4.129999E-05	-3.106718E-06	4.194580E-07
32	-2.913286E-04	4.010179E-05	-1.135333E-06	7.113578E-08
41	-8.422223E-05	1.257386E-05	-6.725352E-07	6.932824E-08
42	2.359834E-04	-3.441824E-05	1.774928E-06	7.075621E-09
51	2.102494E-04	-1.650048E-05	6.376064E-07	-1.018956E-08
52	1.399871E-04	-7.311262E-06	1.469642E-07	-1.307130E-10
61	4.276100E-06	1.573084E-06	-7.263277E-08	6.123500E-11
62	-1.556452E-05	-1.959351E-07	3.379924E-08	-7.683070E-10
71	-3.339924E-05	9.649750E-07	-1.075699E-08	-7.716000E-12
72	-1.917226E-05	8.902803E-07	-2.512078E-08	3.123380E-10
81	2.901655E-05	-9.065185E-07	1.588236E-08	-1.194150E-10
82	1.791836E-05	-6.001143E-07	1.065426E-08	-7.833400E-11

FIG. 31

Seventh Embodiment							
EFL=7.030mm, HFOV=40.845 Degrees, TTL=9.216mm, Fno=1.552, ImgH=5.935mm							
No.		Radius of Curvature (mm)	Aperture Stop Distance/ Thickness/ Air Gap (mm)		Refractive Index	Abbe No.	Focal Length (mm)
	Object	Infinity	Infinity				
2	Ape. Stop	Infinity	-0.905				
11	First Lens element	3.352	1.043	T1	1.545	55.987	7.176
12		20.586	0.252	G12			
21	Second Lens element	8.365	0.341	T2	1.661	20.373	-19.214
22		4.979	0.792	G23			
31	Third Lens element	34.497	0.552	T3	1.661	20.373	-41.205
32		15.201	0.084	G34			
41	Fourth Lens element	-49.628	1.091	T4	1.545	55.987	16.387
42		-7.643	0.585	G45			
51	Fifth Lens element	5.050	0.360	T5	1.661	20.373	-127.655
52		4.631	0.455	G56			
61	Sixth Lens element	7.468	0.679	T6	1.545	55.987	8.984
62		-13.854	0.035	G67			
71	Seventh Lens element	11.437	0.433	T7	1.661	20.373	-34.050
72		7.491	0.981	G78			
81	Eighth Lens element	21.919	0.528	T8	1.545	55.987	-6.200
82		2.909	0.655	G8F			
3	Filter	Infinity	0.210		1.517	64.167	
		Infinity	0.141				
4	Image Plane	Infinity					

FIG. 32

No.	K	a_4	a_6	a_8
11	-4.096711E-02	3.493477E-04	4.426548E-04	-1.808565E-04
12	-6.504531E+00	-3.641085E-03	1.364117E-03	-1.580779E-04
21	-1.303637E+01	-1.298547E-02	5.168751E-03	-7.559868E-04
22	3.823536E+00	-1.715260E-02	4.077465E-03	-5.738924E-04
31	4.968568E+01	-1.636745E-02	-1.775393E-03	8.481401E-04
32	3.449448E+01	-1.120907E-02	-2.003001E-03	9.344988E-04
41	5.003241E+01	1.114580E-03	-8.217334E-04	2.028960E-04
42	-1.093241E+00	-9.639300E-03	2.278934E-03	-9.099348E-04
51	-4.270718E+00	-2.191801E-02	6.193710E-03	-1.533406E-03
52	-1.610359E+00	-2.824456E-02	6.300786E-03	-1.273301E-03
61	2.682056E+00	-4.999650E-03	5.647554E-04	-2.396354E-04
62	-1.653436E+01	2.614984E-02	-5.074984E-03	4.186309E-04
71	-4.065242E+00	1.458271E-02	-5.118834E-03	5.806955E-04
72	-7.159481E+00	4.889168E-03	-2.276339E-03	2.708797E-04
81	1.313831E+01	-4.205675E-02	6.778451E-03	-5.681524E-04
82	-6.776085E+00	-2.063633E-02	3.128968E-03	-3.052488E-04
No.	a_{10}	a_{12}	a_{14}	a_{16}
11	5.162395E-05	-6.054339E-06	1.483418E-07	8.544669E-09
12	-1.211822E-05	4.408055E-06	-3.315139E-07	8.832560E-09
21	6.829163E-06	1.149340E-05	-3.125333E-07	-2.856702E-08
22	-4.426659E-05	1.213369E-05	1.237861E-07	-8.414207E-08
31	-3.044750E-04	4.419556E-05	-1.678507E-06	1.070392E-07
32	-2.968119E-04	4.203503E-05	-9.154805E-07	-6.657104E-08
41	-8.090431E-05	1.196157E-05	-6.332139E-07	4.166937E-08
42	2.381934E-04	-3.319237E-05	1.838613E-06	-1.414419E-08
51	2.127860E-04	-1.657696E-05	6.247548E-07	-7.469675E-09
52	1.400177E-04	-7.312655E-06	1.468494E-07	-1.315090E-10
61	4.600990E-06	1.582335E-06	-7.419448E-08	-2.290000E-12
62	-1.505035E-05	-1.955197E-07	3.288278E-08	-7.328370E-10
71	-3.337786E-05	9.656068E-07	-1.077293E-08	-1.105800E-11
72	-1.916094E-05	8.906749E-07	-2.510739E-08	3.120770E-10
81	2.901700E-05	-9.063667E-07	1.589295E-08	-1.197480E-10
82	1.792091E-05	-6.006396E-07	1.062521E-08	-7.732700E-11

FIG. 33

Embodiment	1st	2nd	3rd	4th	5th	6th	7th
EFL	7.598	8.000	7.038	7.587	7.036	7.031	7.030
Fno	1.677	1.766	1.554	1.675	1.553	1.552	1.552
HFOV	40.842	40.842	40.842	40.842	40.842	40.842	40.845
ImgH	5.793	6.325	6.306	6.133	6.076	5.874	5.935
T1	0.993	1.514	1.065	1.057	0.991	1.034	1.043
G12	0.450	0.073	0.173	0.103	0.252	0.272	0.252
T2	0.270	0.358	0.522	0.267	0.315	0.358	0.341
G23	0.698	0.625	0.755	0.599	0.982	0.764	0.792
T3	0.713	1.027	0.847	0.594	0.384	0.519	0.552
G34	0.024	0.054	0.051	0.273	0.067	0.073	0.084
T4	1.243	0.431	0.424	1.571	1.108	0.957	1.091
G45	0.310	0.330	0.025	0.513	0.371	0.676	0.585
T5	0.336	0.807	0.715	0.419	0.539	0.441	0.360
G56	0.566	0.218	0.322	0.410	0.367	0.392	0.455
T6	0.573	0.874	0.934	0.560	0.839	0.699	0.679
G67	0.033	0.858	0.644	0.976	0.035	0.035	0.035
T7	0.434	0.742	0.704	0.721	0.547	0.466	0.433
G78	0.984	0.809	0.077	0.519	1.331	1.085	0.981
T8	0.326	0.778	1.229	0.578	0.287	0.348	0.528
G8P	0.989	0.253	0.355	0.071	0.655	0.655	0.655
TF	0.210	0.210	0.210	0.210	0.210	0.210	0.210
GFP	0.324	0.309	0.346	0.279	0.129	0.134	0.141
υ1	55.987	55.987	55.987	55.987	55.987	55.987	55.987
υ2	20.373	20.373	20.373	20.373	20.373	20.373	20.373
υ3	19.480	20.373	20.373	20.373	20.373	20.373	20.373
υ4	37.533	55.987	55.987	55.987	55.987	55.987	55.987
υ5	19.480	20.373	20.373	20.373	20.373	20.373	20.373
υ6	49.620	55.987	55.987	55.987	55.987	55.987	55.987
υ7	49.620	20.373	20.373	20.373	20.373	20.373	20.373
υ8	49.620	55.987	55.987	55.987	55.987	55.987	55.987

FIG. 34

Embodiment	1st	2nd	3rd	4th	5th	6th	7th
BFL	1.523	0.772	0.911	0.560	0.994	0.999	1.006
ALT	4.889	6.531	6.439	5.767	5.008	4.822	5.026
AAG	3.065	2.967	2.048	3.393	3.405	3.297	3.183
TL	7.954	9.498	8.487	9.161	8.413	8.119	8.210
TTL	9.477	10.270	9.398	9.721	9.407	9.118	9.216
EPD	4.530	4.530	4.530	4.530	4.530	4.530	4.530
Tavg567	0.448	0.808	0.784	0.567	0.641	0.535	0.491
Tavg5678	0.417	0.800	0.895	0.569	0.553	0.488	0.500
Tmax	1.243	1.514	1.229	1.571	1.108	1.034	1.091
Tmin	0.270	0.358	0.424	0.267	0.287	0.348	0.341
Tstd567	0.097	0.054	0.106	0.123	0.139	0.116	0.136
Tstd5678	0.099	0.048	0.213	0.107	0.195	0.129	0.119
Tavg567/Tstd567	4.600	15.000	7.408	4.600	4.600	4.600	3.602
υ3+υ4	57.013	76.360	76.360	76.360	76.360	76.360	76.360
υ4+υ5	57.013	76.360	76.360	76.360	76.360	76.360	76.360
υ6+υ7	99.240	76.360	76.360	76.360	76.360	76.360	76.360
υ7+υ8	99.240	76.360	76.360	76.360	76.360	76.360	76.360
(EFL+AAG)/BFL	7.000	14.204	9.972	19.612	10.504	10.341	10.149
(TL+ImgH)/D11t32	4.400	4.400	4.400	5.836	4.956	4.749	4.747
(T1+D12t31)/(G34+T4)	1.902	5.300	5.300	1.099	2.161	2.357	2.068
(D12t31+T3)/(G45+T5)	3.300	1.831	3.102	1.676	2.125	1.711	2.050
Tavg5678/Tstd5678	4.200	16.553	4.200	5.333	2.830	3.773	4.199
D52t71/T7	2.700	2.628	2.700	2.700	2.268	2.418	2.697
(D11t32+D52t71)/(T7+G78)	3.030	3.575	6.740	3.684	2.217	2.626	2.933
(D52t71+BFL)/(T7+G78)	1.901	1.754	3.601	2.021	1.190	1.370	1.538
Fno*D11t32/(G78+T8)	4.000	4.000	4.000	4.000	2.806	3.192	3.064
Fno*D52t82/D32t52	2.557	4.658	4.999	2.270	2.536	2.186	2.279
ALT/D32t52	2.556	4.026	5.299	2.077	2.402	2.245	2.372
(EPD+ImgH)/(Tmax+Tmin)	6.823	5.800	6.559	5.800	7.603	7.528	7.311
(TTL+ImgH)/(T1+T3+T8)	7.512	5.000	5.000	7.112	9.320	7.887	7.139

FIG. 35

OPTICAL IMAGING LENS

BACKGROUND OF THE INVENTION

1. Field of the Invention

The present invention generally relates to an optical imaging lens. Specifically speaking, the present invention is directed to an optical imaging lens for use in a mobile electronic device, such as a mobile phone, a camera, a tablet personal computer, or a personal digital assistant (PDA) and for taking pictures or for recording videos.

2. Description of the Prior Art

The specifications of portable electronic devices are changing, and their key components-optical imaging lenses are also developing more diversely. As far as a main lens of a portable electronic device is concerned, it does not only pursues a smaller f-number (Fno) and maintain a shorter system length, but also pursues more pixels and better resolution. More pixels imply the increase of the image height of the lens to receive more imaging rays to meet the pixel demands by using a larger imaging sensor.

However, the design of a small f-number makes the lens receive more imaging rays but more difficult to design. More pixels make the resolution of the lens higher to go with the design of a small f-number to make it much more difficult to design. Therefore, it is a problem to add more lens elements in the limited system length and to increase the resolution while to have a small f-number and a larger image height to be solved.

SUMMARY OF THE INVENTION

In light of the above, the present invention proposes an optical imaging lens of eight lens elements which has a small Fno, a larger image height, improved resolution, maintains good imaging quality and is technically plausible. The optical imaging lens of eight lens elements of the present invention from an object side to an image side in order along an optical axis has a first lens element, a second lens element, a third lens element, a fourth lens element, a fifth lens element, a sixth lens element, a seventh lens element and an eighth lens element. Each of the first lens element, the second lens element, the third lens element, the fourth lens element, the fifth lens element, the sixth lens element, the seventh lens element and the eighth lens element respectively has an object-side surface which faces toward the object side to allow imaging rays to pass through as well as an image-side surface which faces toward the image side to allow the imaging rays to pass through.

In one embodiment of the present invention, the second lens element has negative refracting power, the third lens element has negative refracting power and a periphery region of the object-side surface of the third lens element is concave, the seventh lens element has negative refracting power, an optical axis region of the object-side surface of the eighth lens element is convex and a periphery region of the image-side surface of the eighth lens element is convex. Lens elements included by the optical imaging lens are only the eight lens elements described above to satisfy the relationship: Tavg567/Tstd567≥3.600.

In another embodiment of the present invention, the second lens element has negative refracting power, a periphery region of the object-side surface of the third lens element is concave, an optical axis region of the object-side surface of the fifth lens element is convex and a periphery region of the object-side surface of the fifth lens element is concave, an optical axis region of the object-side surface of the sixth lens element is convex, the seventh lens element has negative refracting power, and a periphery region of the image-side surface of the eighth lens element is convex. Lens elements included by the optical imaging lens are only the eight lens elements described above to satisfy the relationship: Tavg567/Tstd567≥4.600.

In still another embodiment of the present invention, the first lens element has positive refracting power, the second lens element has negative refracting power, a periphery region of the object-side surface of the third lens element is concave, an optical axis region of the object-side surface of the fifth lens element is convex and a periphery region of the image-side surface of the fifth lens element is convex, an optical axis region of the object-side surface of the sixth lens element is convex, the seventh lens element has negative refracting power, and a periphery region of the image-side surface of the eighth lens element is convex. Lens elements included by the optical imaging lens are only the eight lens elements described above, to satisfy the relationship: Tavg567/Tstd567≥4.600.

In the optical imaging lens of the present invention, each embodiment may also selectively satisfy the following numerical conditions:

$$55.000 \leq \upsilon3+\upsilon4;$$

$$\upsilon7+\upsilon8 \leq 100.000;$$

$$(T1+D12t31)/(G34+T4) \leq 5.300;$$

$$D52t71/T7 \leq 2.700;$$

$$Fno*D11t32/(G78+T8) \leq 4.000;$$

$$5.800 \leq (EPD+ImgH)/(T\max+T\min);$$

$$55.000 \leq \upsilon4+\upsilon5;$$

$$7.000 \leq (EFL+AAG)/BFL;$$

$$(D12t31+T3)/(G45+T5) \leq 3.300;$$

$$(D11t32+D52t71)/(T7+G78) \leq 7.000;$$

$$Fno*D52t82/D32t52 \leq 5.000;$$

$$5.000 \leq (TTL+ImgH)/(T1+T3+T8);$$

$$\upsilon6+\upsilon7 \leq 100.000;$$

$$4.400 \leq (TL+ImgH)/D11t32;$$

$$2.800 \leq T\text{avg}5678/\text{Tstd}5678;$$

$$(D52t71+BFL)/(T7+G78) \leq 3.610;$$

$$ALT/D32t52 \leq 5.300.$$

In the present invention, υ3 is an Abbe number of the third lens element, υ4 is an Abbe number of the fourth lens element, υ5 is an Abbe number of the fifth lens element, υ6 is an Abbe number of the sixth lens element, υ7 is an Abbe number of the seventh lens element, υ8 is an Abbe number of the eighth lens element, T1 is a thickness of the first lens element along the optical axis, T3 is a thickness of the third lens element along the optical axis, T4 is a thickness of the fourth lens element along the optical axis, T5 is a thickness of the fifth lens element along the optical axis, T7 is a thickness of the seventh lens element along the optical axis, and T8 is a thickness of the eighth lens element along the optical axis. Tmax is a maximal lens element thickness from the first lens element to the eighth lens element along the optical axis, namely the maximum among T1, T2, T3, T4, T5, T6, T7 and T8. Tmin is a minimal lens element thickness from the first lens element to the eighth lens element along the optical axis, namely the minimum among T1, T2, T3, T4, T5, T6, T7 and T8. G34 is an air gap between the third lens element and the fourth lens element along the optical axis, G45 is an air gap between the fourth lens element and the fifth lens element along the optical axis, G78 is an air gap between the seventh lens element and the eighth lens element along the optical axis.

Further, D11t32 is defined as a distance from the object-side surface of the first lens element to the image-side surface of the third lens element along the optical axis, D12t31 is defined as a distance from the image-side surface of the first lens element to the object-side surface of the third lens element along the optical axis, D52t71 is defined as a distance from the image-side surface of the fifth lens element to the object-side surface of the seventh lens element along the optical axis, D52t82 is defined as a distance from the image-side surface of the fifth lens element to the image-side surface of the eighth lens element along the optical axis, D32t52 is defined as a distance from the image-side surface of the third lens element to the image-side surface of the fifth lens element along the optical axis. Tavg567 is an average of three thicknesses from the fifth lens element to the seventh lens element along the optical axis, namely the average of T5, T6 and T7. Tstd567 is a population standard deviation of the three thicknesses from the fifth lens element to the seventh lens element along the optical axis, namely the population standard deviation of T5, T6 and T7. Tavg5678 is an average of four thicknesses from the fifth lens element to the eighth lens element along the optical axis, namely the average of T5, T6, T7 and T8. Tstd5678 is a population standard deviation of the four thicknesses from the fifth lens element to the eighth lens element along the optical axis, namely the population standard deviation of T5, T6, T7 and T8.

TTL is a distance from the object-side surface of the first lens element to an image plane along the optical axis, ALT is a sum of the thicknesses of eight lens elements from the first lens element to the eighth lens element along the optical axis, Tavg is an average of the eight thicknesses from the first lens element to the eighth lens element along the optical axis, EPD is an entrance pupil diameter of the optical imaging lens, TL is a distance from the object-side surface of the first lens element to the image-side surface of the eighth lens element along the optical axis, AAG is a sum of seven air gaps from the first lens element to the eighth lens element along the optical axis, EFL is an effective focal length of the optical imaging lens, Fno is a f-number of the optical imaging lens, BFL is a distance from the image-side surface of the eighth lens element to the image plane along the optical axis, and ImgH is an image height of the optical imaging lens.

These and other objectives of the present invention will no doubt become obvious to those of ordinary skill in the art after reading the following detailed description of the preferred embodiment that is illustrated in the various figures and drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

FIGS. **1-5** illustrate the methods for determining the surface shapes and for determining an optical axis region and a periphery region of one lens element.

FIG. **6** illustrates a first embodiment of the optical imaging lens of the present invention.

FIG. **7**A illustrates the longitudinal spherical aberration on the image plane of the first embodiment.

FIG. **7**B illustrates the field curvature aberration on the sagittal direction of the first embodiment.

FIG. **7**C illustrates the field curvature aberration on the tangential direction of the first embodiment.

FIG. **7**D illustrates the distortion of the first embodiment.

FIG. **8** illustrates a second embodiment of the optical imaging lens of the present invention.

FIG. **9**A illustrates the longitudinal spherical aberration on the image plane of the second embodiment.

FIG. **9**B illustrates the field curvature aberration on the sagittal direction of the second embodiment.

FIG. **9**C illustrates the field curvature aberration on the tangential direction of the second embodiment.

FIG. **9**D illustrates the distortion of the second embodiment.

FIG. **10** illustrates a third embodiment of the optical imaging lens of the present invention.

FIG. **11**A illustrates the longitudinal spherical aberration on the image plane of the third embodiment.

FIG. **11**B illustrates the field curvature aberration on the sagittal direction of the third embodiment.

FIG. **11**C illustrates the field curvature aberration on the tangential direction of the third embodiment.

FIG. **11**D illustrates the distortion of the third embodiment.

FIG. **12** illustrates a fourth embodiment of the optical imaging lens of the present invention.

FIG. **13**A illustrates the longitudinal spherical aberration on the image plane of the fourth embodiment.

FIG. **13**B illustrates the field curvature aberration on the sagittal direction of the fourth embodiment.

FIG. **13**C illustrates the field curvature aberration on the tangential direction of the fourth embodiment.

FIG. **13**D illustrates the distortion of the fourth embodiment.

FIG. **14** illustrates a fifth embodiment of the optical imaging lens of the present invention.

FIG. **15**A illustrates the longitudinal spherical aberration on the image plane of the fifth embodiment.

FIG. **15**B illustrates the field curvature aberration on the sagittal direction of the fifth embodiment.

FIG. **15**C illustrates the field curvature aberration on the tangential direction of the fifth embodiment.

FIG. **15**D illustrates the distortion of the fifth embodiment.

FIG. **16** illustrates a sixth embodiment of the optical imaging lens of the present invention.

FIG. **17**A illustrates the longitudinal spherical aberration on the image plane of the sixth embodiment.

FIG. **17**B illustrates the field curvature aberration on the sagittal direction of the sixth embodiment.

FIG. **17**C illustrates the field curvature aberration on the tangential direction of the sixth embodiment.

FIG. **17**D illustrates the distortion of the sixth embodiment.

FIG. **18** illustrates a seventh embodiment of the optical imaging lens of the present invention.

FIG. **19**A illustrates the longitudinal spherical aberration on the image plane of the seventh embodiment.

FIG. **19**B illustrates the field curvature aberration on the sagittal direction of the seventh embodiment.

FIG. **19**C illustrates the field curvature aberration on the tangential direction of the seventh embodiment.

FIG. 19D illustrates the distortion of the seventh embodiment.

FIG. 20 shows the optical data of the first embodiment of the optical imaging lens.

FIG. 21 shows the aspheric surface data of the first embodiment.

FIG. 22 shows the optical data of the second embodiment of the optical imaging lens.

FIG. 23 shows the aspheric surface data of the second embodiment.

FIG. 24 shows the optical data of the third embodiment of the optical imaging lens.

FIG. 25 shows the aspheric surface data of the third embodiment.

FIG. 26 shows the optical data of the fourth embodiment of the optical imaging lens.

FIG. 27 shows the aspheric surface data of the fourth embodiment.

FIG. 28 shows the optical data of the fifth embodiment of the optical imaging lens.

FIG. 29 shows the aspheric surface data of the fifth embodiment.

FIG. 30 shows the optical data of the sixth embodiment of the optical imaging lens.

FIG. 31 shows the aspheric surface data of the sixth embodiment.

FIG. 32 shows the optical data of the seventh embodiment of the optical imaging lens.

FIG. 33 shows the aspheric surface data of the seventh embodiment.

FIG. 34 shows some important parameters in the embodiments.

FIG. 35 shows some important ratios in the embodiments.

DETAILED DESCRIPTION

The terms "optical axis region", "periphery region", "concave", and "convex" used in this specification and claims should be interpreted based on the definition listed in the specification by the principle of lexicographer.

In the present disclosure, the optical system may comprise at least one lens element to receive imaging rays that are incident on the optical system over a set of angles ranging from parallel to an optical axis to a half field of view (HFOV) angle with respect to the optical axis. The imaging rays pass through the optical system to produce an image on an image plane. The term "a lens element having positive refracting power (or negative refracting power)" means that the paraxial refracting power of the lens element in Gaussian optics is positive (or negative). The term "an object-side (or image-side) surface of a lens element" refers to a specific region of that surface of the lens element at which imaging rays can pass through that specific region. Imaging rays include at least two types of rays: a chief ray Lc and a marginal ray Lm (as shown in FIG. 1). An object-side (or image-side) surface of a lens element can be characterized as having several regions, including an optical axis region, a periphery region, and, in some cases, one or more intermediate regions, as discussed more fully below.

FIG. 1 is a radial cross-sectional view of a lens element 100. Two referential points for the surfaces of the lens element 100 can be defined: a central point, and a transition point. The central point of a surface of a lens element is a point of intersection of that surface and the optical axis I. As illustrated in FIG. 1, a first central point CP1 may be present on the object-side surface 110 of lens element 100 and a second central point CP2 may be present on the image-side surface 120 of the lens element 100. The transition point is a point on a surface of a lens element, at which the line tangent to that point is perpendicular to the optical axis I. The optical boundary OB of a surface of the lens element is defined as a point at which the radially outermost marginal ray Lm passing through the surface of the lens element intersects the surface of the lens element. All transition points lie between the optical axis I and the optical boundary OB of the surface of the lens element. A surface of the lens element 100 may have no transition point or have at least one transition point. If multiple transition points are present on a single surface, then these transition points are sequentially named along the radial direction of the surface with reference numerals starting from the first transition point. For example, the first transition point, e.g., TP1, (closest to the optical axis I), the second transition point, e.g., TP2, (as shown in FIG. 4), and the Nth transition point (farthest from the optical axis I).

When a surface of the lens element has at least one transition point, the region of the surface of the lens element from the central point to the first transition point TP1 is defined as the optical axis region, which includes the central point. The region located radially outside of the farthest transition point (the Nth transition point) from the optical axis I to the optical boundary OB of the surface of the lens element is defined as the periphery region. In some embodiments, there may be intermediate regions present between the optical axis region and the periphery region, with the number of intermediate regions depending on the number of the transition points. When a surface of the lens element has no transition point, the optical axis region is defined as a region of 0%-50% of the distance between the optical axis I and the optical boundary OB of the surface of the lens element, and the periphery region is defined as a region of 50%-100% of the distance between the optical axis I and the optical boundary OB of the surface of the lens element.

The shape of a region is convex if a collimated ray being parallel to the optical axis I and passing through the region is bent toward the optical axis I such that the ray intersects the optical axis I on the image side A2 of the lens element. The shape of a region is concave if the extension line of a collimated ray being parallel to the optical axis I and passing through the region intersects the optical axis I on the object side A1 of the lens element.

Additionally, referring to FIG. 1, the lens element 100 may also have a mounting portion 130 extending radially outward from the optical boundary OB. The mounting portion 130 is typically used to physically secure the lens element to a corresponding element of the optical system (not shown). Imaging rays do not reach the mounting portion 130. The structure and shape of the mounting portion 130 are only examples to explain the technologies, and should not be taken as limiting the scope of the present disclosure. The mounting portion 130 of the lens elements discussed below may be partially or completely omitted in the following drawings.

Referring to FIG. 2, optical axis region Z1 is defined between central point CP and first transition point TP1. Periphery region Z2 is defined between TP1 and the optical boundary OB of the surface of the lens element. Collimated ray 211 intersects the optical axis Ion the image side A2 of lens element 200 after passing through optical axis region Z1, i.e., the focal point of collimated ray 211 after passing through optical axis region Z1 is on the image side A2 of the lens element 200 at point R in FIG. 2. Accordingly, since the ray itself intersects the optical axis I on the image side A2 of the lens element 200, optical axis region Z1 is convex. On the contrary, collimated ray **212** diverges after passing through periphery region Z2. The extension line EL of collimated ray **212** after passing through periphery region Z2 intersects the optical axis I on the object side A1 of lens element **200**, i.e., the focal point of collimated ray **212** after passing through periphery region Z2 is on the object side A1 at point M in FIG. **2**. Accordingly, since the extension line EL of the ray intersects the optical axis I on the object side A1 of the lens element **200**, periphery region Z2 is concave. In the lens element **200** illustrated in FIG. **2**, the first transition point TP1 is the border of the optical axis region and the periphery region, i.e., TP1 is the point at which the shape changes from convex to concave.

Alternatively, there is another way for a person having ordinary skill in the art to determine whether an optical axis region is convex or concave by referring to the sign of "Radius of curvature" (the "R" value), which is the paraxial radius of shape of a lens surface in the optical axis region. The R value is commonly used in conventional optical design software such as Zemax and CodeV. The R value usually appears in the lens data sheet in the software. For an object-side surface, a positive R value defines that the optical axis region of the object-side surface is convex, and a negative R value defines that the optical axis region of the object-side surface is concave. Conversely, for an image-side surface, a positive R value defines that the optical axis region of the image-side surface is concave, and a negative R value defines that the optical axis region of the image-side surface is convex. The result found by using this method should be consistent with the method utilizing intersection of the optical axis by rays/extension lines mentioned above, which determines surface shape by referring to whether the focal point of a collimated ray being parallel to the optical axis I is on the object-side or the image-side of a lens element. As used herein, the terms "a shape of a region is convex (concave)," "a region is convex (concave)," and "a convex-(concave-) region," can be used alternatively.

FIG. **3**, FIG. **4** and FIG. **5** illustrate examples of determining the shape of lens element regions and the boundaries of regions under various circumstances, including the optical axis region, the periphery region, and intermediate regions as set forth in the present specification.

FIG. **3** is a radial cross-sectional view of a lens element **300**. As illustrated in FIG. **3**, only one transition point TP1 appears within the optical boundary OB of the image-side surface **320** of the lens element **300**. Optical axis region Z1 and periphery region Z2 of the image-side surface **320** of lens element **300** are illustrated. The R value of the image-side surface **320** is positive (i.e., R>0). Accordingly, the optical axis region Z1 is concave.

In general, the shape of each region demarcated by the transition point will have an opposite shape to the shape of the adjacent region(s). Accordingly, the transition point will define a transition in shape, changing from concave to convex at the transition point or changing from convex to concave. In FIG. **3**, since the shape of the optical axis region Z1 is concave, the shape of the periphery region Z2 will be convex as the shape changes at the transition point TP1.

FIG. **4** is a radial cross-sectional view of a lens element **400**. Referring to FIG. **4**, a first transition point TP1 and a second transition point TP2 are present on the object-side surface **410** of lens element **400**. The optical axis region Z1 of the object-side surface **410** is defined between the optical axis I and the first transition point TP1. The R value of the object-side surface **410** is positive (i.e., R>0). Accordingly, the optical axis region Z1 is convex.

The periphery region Z2 of the object-side surface **410**, which is also convex, is defined between the second transition point TP2 and the optical boundary OB of the object-side surface **410** of the lens element **400**. Further, intermediate region Z3 of the object-side surface **410**, which is concave, is defined between the first transition point TP1 and the second transition point TP2. Referring once again to FIG. **4**, the object-side surface **410** includes an optical axis region Z1 located between the optical axis I and the first transition point TP1, an intermediate region Z3 located between the first transition point TP1 and the second transition point TP2, and a periphery region Z2 located between the second transition point TP2 and the optical boundary OB of the object-side surface **410**. Since the shape of the optical axis region Z1 is designed to be convex, the shape of the intermediate region Z3 is concave as the shape of the intermediate region Z3 changes at the first transition point TP1, and the shape of the periphery region Z2 is convex as the shape of the periphery region Z2 changes at the second transition point TP2.

FIG. **5** is a radial cross-sectional view of a lens element **500**. Lens element **500** has no transition point on the object-side surface **510** of the lens element **500**. For a surface of a lens element with no transition point, for example, the object-side surface **510** the lens element **500**, the optical axis region Z1 is defined as the region of 0%-50% of the distance between the optical axis I and the optical boundary OB of the surface of the lens element and the periphery region is defined as the region of 50%-100% of the distance between the optical axis I and the optical boundary OB of the surface of the lens element. Referring to lens element **500** illustrated in FIG. **5**, the optical axis region Z1 of the object-side surface **510** is defined between the optical axis I and 50% of the distance between the optical axis I and the optical boundary OB. The R value of the object-side surface **510** is positive (i.e., R>0). Accordingly, the optical axis region Z1 is convex. For the object-side surface **510** of the lens element **500**, because there is no transition point, the periphery region Z2 of the object-side surface **510** is also convex. It should be noted that lens element **500** may have a mounting portion (not shown) extending radially outward from the periphery region Z2.

As shown in FIG. **6**, the optical imaging lens **1** of eight lens elements of the present invention, sequentially located from an object side A1 (where an object is located) to an image side A2 along an optical axis I, has a first lens element **10**, a second lens element **20**, a third lens element **30**, a fourth lens element **40**, a fifth lens element **50**, a sixth lens element **60**, a seventh lens element **70**, an eighth lens element **80**, and an image plane **4**. Generally speaking, the first lens element **10**, the second lens element **20**, the third lens element **30**, the fourth lens element **40**, the fifth lens element **50**, the sixth lens element **60**, the seventh lens element **70** and the eighth lens element **80** may be made of a transparent plastic material but the present invention is not limited to this. In the present invention, lens elements included by the optical imaging lens **1** are only the eight lens elements, for example the first lens element **10**, the second lens element **20**, the third lens element **30**, the fourth lens element **40**, the fifth lens element **50**, the sixth lens element **60**, the seventh lens element **70** and the eighth lens element **80**, described above. The optical axis I is the optical axis of the entire optical imaging lens **1**, and the optical axis of each of the lens elements coincides with the optical axis of the optical imaging lens **1**.

Furthermore, the optical imaging lens **1** includes an aperture stop (ape. stop) 2 disposed in an appropriate position. In FIG. **6**, the aperture stop **2** is disposed at a side of the first lens element **10** facing the object side A1. In other words, the first lens element **10** is disposed between the aperture stop **2** and the second lens element **20**. When light emitted or reflected by an object (not shown) which is located at the object side A1 enters the optical imaging lens **1** of the present invention, it forms a clear and sharp image on the image plane **4** at the image side A2 after passing through the aperture stop **2**, the first lens element **10**, the second lens element **20**, the third lens element **30**, the fourth lens element **40**, the fifth lens element **50**, the sixth lens element **60**, the seventh lens element **70**, the eighth lens element **80** and the filter **3**. In each embodiment of the present invention, placed between the eighth lens element **80** and the image plane **4** to filter out light of a specific wavelength, for some embodiments, the optional filter **3** may be a filter of various suitable functions, for example, the filter **3** may be an infrared cut filter (IR cut filter) to keep infrared light in the imaging rays from reaching the image plane **4** to jeopardize the imaging quality.

Each lens element in the optical imaging lens **1** of the present invention has an object-side surface facing toward the object side A1 as well as an image-side surface facing toward the image side A2, and each lens element also has an optical axis region and a periphery region respectively. For example, the first lens element **10** has an object-side surface **11** and an image-side surface **12**; the second lens element **20** has an object-side surface **21** and an image-side surface **22**; the third lens element **30** has an object-side surface **31** and an image-side surface **32**; the fourth lens element **40** has an object-side surface **41** and an image-side surface **42**; the fifth lens element **50** has an object-side surface **51** and an image-side surface **52**; the sixth lens element **60** has an object-side surface **61** and an image-side surface **62**; the seventh lens element **70** has an object-side surface **71** and an image-side surface **72**; and the eighth lens element **80** has an object-side surface **81** and an image-side surface **82**. In addition, each object-side surface and image-side surface in the optical imaging lens **1** of the present invention has an optical axis region and a periphery region.

Each lens element in the optical imaging lens **1** of the present invention further has a thickness T along the optical axis I. For example, the first lens element **10** has a first lens element thickness T1, the second lens element **20** has a second lens element thickness T2, the third lens element **30** has a third lens element thickness T3, the fourth lens element **40** has a fourth lens element thickness T4, the fifth lens element **50** has a fifth lens element thickness T5, the sixth lens element **60** has a sixth lens element thickness T6, the seventh lens element **70** has a seventh lens element thickness T7, the eighth lens element **80** has an eighth lens element thickness T8. Therefore, a sum of the thicknesses of eight lens elements from the first lens element **10** to the eighth lens element **80** in the optical imaging lens **1** along the optical axis I is ALT=T1+T2+T3+T4+T5+T6+T7+T8.

Tmax is a maximal lens element thickness from the first lens element **10** to the eighth lens element **80** along the optical axis I, namely the maximum among T1, T2, T3, T4, T5, T6, T7 and T8. Tmin is a minimal lens element thickness from the first lens element **10** to the eighth lens element **80** along the optical axis I, namely the minimum among T1, T2, T3, T4, T5, T6, T7 and T8. Tavg is an average of the eight thicknesses from the first lens element **10** to the eighth lens element **80** along the optical axis I, that is Tavg=ALT/8. Tavg567 is an average of three thicknesses from the fifth lens element **50** to the seventh lens element **70** along the optical axis, namely the average of T5, T6 and T7. Tstd567 is a population standard deviation of the three thicknesses from the fifth lens element **50** to the seventh lens element **70** along the optical axis I, namely the population standard deviation of T5, T6 and T7. Tavg5678 is an average of four thicknesses from the fifth lens element **50** to the eighth lens element **80** along the optical axis I, namely the average of T5, T6, T7 and T8. Tstd5678 is a population standard deviation of the four thicknesses from the fifth lens element **50** to the eighth lens element **80** along the optical axis I, namely the population standard deviation of T5, T6, T7 and T8.

In addition, between two adjacent lens elements in the optical imaging lens **1** of the present invention there may be an air gap along the optical axis I. For example, there is an air gap G12 disposed between the first lens element **10** and the second lens element **20**, an air gap G23 disposed between the second lens element **20** and the third lens element **30**, an air gap G34 disposed between the third lens element **30** and the fourth lens element **40**, an air gap G45 disposed between the fourth lens element **40** and the fifth lens element **50**, an air gap G56 disposed between the fifth lens element **50** and the sixth lens element **60**, an air gap G67 disposed between the sixth lens element **60** and the seventh lens element **70** as well as an air gap G78 disposed between the seventh lens element **70** and the eighth lens element **80**. Therefore, a sum of seven air gaps from the first lens element **10** to the eighth lens element **80** along the optical axis I is AAG=G12+G23+G34+G45+G56+G67+G78.

Further, D11t32 is a distance from the object-side surface **11** of the first lens element **10** to the image-side surface **32** of the third lens element **30** along the optical axis I, D12t31 is a distance from the image-side surface **12** of the first lens element **10** to the object-side surface **31** of the third lens element **30** along the optical axis I, D52t71 is a distance from the image-side surface **52** of the fifth lens element **50** to the object-side surface **71** of the seventh lens element **70** along the optical axis I, D52t82 is a distance from the image-side surface **52** of the fifth lens element **50** to the image-side surface **82** of the eighth lens element **80** along the optical axis I, D32t52 is a distance from the image-side surface **32** of the third lens element **30** to the image-side surface **52** of the fifth lens element **50** along the optical axis I.

In addition, a distance from the object-side surface **11** of the first lens element **10** to the image plane **4** along the optical axis I is TTL, namely a system length of the optical imaging lens **1**; an effective focal length of the optical imaging lens element is EFL; a distance from the object-side surface **11** of the first lens element **10** to the image-side surface **82** of the eighth lens element **80** along the optical axis I is TL; HFOV stands for the half field of view which is half of the field of view of the entire optical imaging lens; ImgH is an image height of the optical imaging lens **1**, and Fno is a f-number of the optical imaging lens **1**.

When the filter **3** is placed between the eighth lens element **80** and the image plane **4**, an air gap between the eighth lens element **80** and the filter **3** along the optical axis I is G8F; a thickness of the filter **3** along the optical axis I is TF; an air gap between the filter **3** and the image plane **4** along the optical axis I is GFP; and a distance from the image-side surface **82** of the eighth lens element **80** to the image plane **4** along the optical axis I is BFL. Therefore, BFL=G8F+TF+GFP.

Furthermore, a focal length of the first lens element **10** is f1; a focal length of the second lens element **20** is f2; a focal length of the third lens element **30** is f3; a focal length of the fourth lens element **40** is f4; a focal length of the fifth lens element **50** is f5; a focal length of the sixth lens element **60** is f6; a focal length of the seventh lens element **70** is f7; a focal length of the eighth lens element **80** is f8; a refractive index of the first lens element **10** is n1; a refractive index of the second lens element **20** is n2; a refractive index of the third lens element **30** is n3; a refractive index of the fourth lens element **40** is n4; a refractive index of the fifth lens element **50** is n5; a refractive index of the sixth lens element **60** is n6; a refractive index of the seventh lens element **70** is n7; a refractive index of the eighth lens element **80** is n8; an Abbe number of the first lens element **10** is υ1; an Abbe number of the second lens element **20** is υ2; an Abbe number of the third lens element **30** is υ3; and an Abbe number of the fourth lens element **40** is υ4; an Abbe number of the fifth lens element **50** is υ5; an Abbe number of the sixth lens element **60** is υ6; an Abbe number of the seventh lens element **70** is υ7; and an Abbe number of the eighth lens element **80** is υ8.

First Embodiment

Please refer to FIG. **6** which illustrates the first embodiment of the optical imaging lens **1** of the present invention. Please refer to FIG. **7**A for the longitudinal spherical aberration on the image plane **4** of the first embodiment; please refer to FIG. **7**B for the field curvature aberration on the sagittal direction; please refer to FIG. **7**C for the field curvature aberration on the tangential direction; and please refer to FIG. **7**D for the distortion aberration. The Y axis of the spherical aberration in each embodiment is "field of view" for 1.0. The Y axis of the astigmatic field and the distortion in each embodiment stands for "image height" (ImgH), which is 5.793 mm.

The optical imaging lens **1** of the first embodiment has eight lens elements with refracting power, an aperture stop **2** and an image plane **4**. The aperture stop **2** is provided at a side of the first lens element **10** facing the object side A1.

The first lens element **10** has positive refracting power. An optical axis region **13** of the object-side surface **11** of the first lens element **10** is convex, and a periphery region **14** of the object-side surface **11** of the first lens element **10** is convex. An optical axis region **16** of the image-side surface **12** of the first lens element **10** is concave, and a periphery region **17** of the image-side surface **12** of the first lens element **10** is concave. Besides, both the object-side surface **11** and the image-side surface **12** of the first lens element **10** are aspheric surfaces, but it is not limited thereto.

The second lens element **20** has negative refracting power. An optical axis region **23** of the object-side surface **21** of the second lens element **20** is convex, and a periphery region **24** of the object-side surface **21** of the second lens element **20** is convex. An optical axis region **26** of the image-side surface **22** of the second lens element **20** is concave, and a periphery region **27** of the image-side surface **22** of the second lens element **20** is concave. Besides, both the object-side surface **21** and the image-side surface **22** of the second lens element **20** are aspheric surfaces, but it is not limited thereto.

The third lens element **30** has negative refracting power. An optical axis region **33** of the object-side surface **31** of the third lens element **30** is concave, and a periphery region **34** of the object-side surface **31** of the third lens element **30** is concave. An optical axis region **36** of the image-side surface **32** of the third lens element **30** is concave, and a periphery region **37** of the image-side surface **32** of the third lens element **30** is convex. Besides, both the object-side surface **31** and the image-side surface **32** of the third lens element **30** are aspheric surfaces, but it is not limited thereto.

The fourth lens element **40** has positive refracting power. An optical axis region **43** of the object-side surface **41** of the fourth lens element **40** is convex, and a periphery region **44** of the object-side surface **41** of the fourth lens element **40** is concave. An optical axis region **46** of the image-side surface **42** of the fourth lens element **40** is convex, and a periphery region **47** of the image-side surface **42** of the fourth lens element **40** is convex. Besides, both the object-side surface **41** and the image-side surface **42** of the fourth lens element **40** are aspheric surfaces, but it is not limited thereto.

The fifth lens element **50** has negative refracting power. An optical axis region **53** of the object-side surface **51** of the fifth lens element **50** is convex, and a periphery region **54** of the object-side surface **51** of the fifth lens element **50** is concave. An optical axis region **56** of the image-side surface **52** of the fifth lens element **50** is concave, and a periphery region **57** of the image-side surface **52** of the fifth lens element **50** is convex. Besides, both the object-side surface **51** and the image-side surface **52** of the fifth lens element **50** are aspheric surfaces, but it is not limited thereto.

The sixth lens element **60** has positive refracting power. An optical axis region **63** of the object-side surface **61** of the sixth lens element **60** is convex, and a periphery region **64** of the object-side surface **61** of the sixth lens element **60** is concave. An optical axis region **66** of the image-side surface **62** of the sixth lens element **60** is convex, and a periphery region **67** of the image-side surface **62** of the sixth lens element **60** is convex. Besides, both the object-side surface **61** and the image-side surface **62** of the sixth lens element **60** are aspheric surfaces, but it is not limited thereto.

The seventh lens element **70** has negative refracting power. An optical axis region **73** of the object-side surface **71** of the seventh lens element **70** is convex, and a periphery region **74** of the object-side surface **71** of the seventh lens element **70** is concave. An optical axis region **76** of the image-side surface **72** of the seventh lens element **70** is concave, and a periphery region **77** of the image-side surface **72** of the seventh lens element **70** is convex. Besides, both the object-side surface **71** and the image-side surface **72** of the seventh lens element **70** are aspheric surfaces, but it is not limited thereto.

The eighth lens element **80** has negative refracting power. An optical axis region **83** of the object-side surface **81** of the eighth lens element **80** is convex, and a periphery region **84** of the object-side surface **81** of the eighth lens element **80** is convex. An optical axis region **86** of the image-side surface **82** of the eighth lens element **80** is concave, and a periphery region **87** of the image-side surface **82** of the eighth lens element **80** is convex. Besides, both the object-side surface **81** and the image-side surface **82** of the eighth lens element **80** are aspheric surfaces, but it is not limited thereto.

In the first lens element **10**, the second lens element **20**, the third lens element **30**, the fourth lens element **40**, the fifth lens element **50**, the sixth lens element **60**, the seventh lens element **70** and the eighth lens element **80** of the optical imaging lens element **1** of the present invention, there are 16 surfaces, such as the object-side surfaces **11/21/31/41/51/61/71/81** and the image-side surfaces **12/22/32/42/52/62/72/82**. If a surface is aspheric, these aspheric coefficients are defined according to the following formula:

$$Z(Y)=\frac{Y^2}{R}\Bigg/\left(1+\sqrt{1-(1+K)\frac{Y^2}{R^2}}\right)+\sum_{i=1}^{n}a_i\times Y^i$$

In which:

Y represents a vertical distance from a point on the aspheric surface to the optical axis;

Z represents the depth of an aspheric surface (the perpendicular distance between the point of the aspheric surface at a distance Y from the optical axis and the tangent plane of the vertex on the optical axis of the aspheric surface);

R represents the radius of curvature of the lens element surface;

K is a conic constant; and

a_i, is the aspheric coefficient of the i^{th} order, and the a_2 coefficient in each embodiment is 0.

The optical data of the first embodiment of the optical imaging lens **1** are shown in FIG. **20** while the aspheric surface data are shown in FIG. **21**. In the present embodiments of the optical imaging lens, the f-number of the entire optical imaging lens element system is Fno, EFL is the effective focal length, HFOV stands for the half field of view which is half of the field of view of the entire optical imaging lens, and the unit for the radius of curvature, the thickness and the focal length is in millimeters (mm). In this embodiment, EFL=7.598 mm; HFOV=40.842 degrees; TTL=9.477 mm; Fno=1.677; ImgH=5.793 mm.

Second Embodiment

Please refer to FIG. **8** which illustrates the second embodiment of the optical imaging lens **1** of the present invention. It is noted that from the second embodiment to the following embodiments, in order to simplify the figures, only the components different from what the first embodiment has, and the basic lens elements will be labeled in figures. Other components that are the same as what the first embodiment has, such as the object-side surface, the image-side surface, the optical axis region and the periphery region will be omitted in the following embodiments. Please refer to FIG. **9**A for the longitudinal spherical aberration on the image plane **4** of the second embodiment, please refer to FIG. **9**B for the field curvature aberration on the sagittal direction, please refer to FIG. **9**C for the field curvature aberration on the tangential direction, and please refer to FIG. **9**D for the distortion aberration. The components in this embodiment are similar to those in the first embodiment, but the optical data such as the radius of curvature, the lens thickness, the aspheric surface or the back focal length in this embodiment are different from the optical data in the first embodiment. Besides, in this embodiment, the optical axis region **36** of the image-side surface **32** of the third lens element **30** is convex, the optical axis region **73** of the object-side surface **71** of the seventh lens element **70** is concave, and the periphery region **84** of the object-side surface **81** of the eighth lens element **80** is concave.

The optical data of the second embodiment of the optical imaging lens are shown in FIG. **22** while the aspheric surface data are shown in FIG. **23**. In this embodiment, EFL=8.000 mm; HFOV=40.842 degrees; TTL=10.270 mm; Fno=1.766; ImgH=6.325 mm. In particular: 1. the longitudinal spherical aberration in this embodiment is better than the longitudinal spherical aberration in the first embodiment; 2. the distortion aberration in this embodiment is better than the distortion aberration in the first embodiment.

Third Embodiment

Please refer to FIG. **10** which illustrates the third embodiment of the optical imaging lens **1** of the present invention. Please refer to FIG. **11**A for the longitudinal spherical aberration on the image plane **4** of the third embodiment; please refer to FIG. **11**B for the field curvature aberration on the sagittal direction; please refer to FIG. **11**C for the field curvature aberration on the tangential direction; and please refer to FIG. **11**D for the distortion aberration. The components in this embodiment are similar to those in the first embodiment, but the optical data such as the radius of curvature, the lens thickness, the aspheric surface or the back focal length in this embodiment are different from the optical data in the first embodiment. Besides, in this embodiment, the fifth lens element **50** has positive refracting power, the optical axis region **76** of the image-side surface **72** of the seventh lens element **70** is convex, and a periphery region **77** of the image-side surface **72** of the seventh lens element **70** is concave.

The optical data of the third embodiment of the optical imaging lens are shown in FIG. **24** while the aspheric surface data are shown in FIG. **25**. In this embodiment, EFL=7.038 mm; HFOV=40.842 degrees; TTL=9.398 mm; Fno=1.554; ImgH=6.306 mm. In particular: 1. the system length of the optical imaging lens TTL in this embodiment is shorter than the system length of the optical imaging lens TTL in the first embodiment; 2. the Fno in this embodiment is smaller than the Fno in the first embodiment; 3. the distortion aberration in this embodiment is better than the distortion aberration in the first embodiment; 4. the thickness ratio of the optical axis region to the periphery region in this embodiment is smaller than that of the optical imaging lens in the first embodiment so it is easier to fabricate to have better yield.

Fourth Embodiment

Please refer to FIG. **12** which illustrates the fourth embodiment of the optical imaging lens **1** of the present invention. Please refer to FIG. **13**A for the longitudinal spherical aberration on the image plane **4** of the fourth embodiment; please refer to FIG. **13**B for the field curvature aberration on the sagittal direction; please refer to FIG. **13**C for the field curvature aberration on the tangential direction; and please refer to FIG. **13**D for the distortion aberration. The components in this embodiment are similar to those in the first embodiment, but the optical data such as the radius of curvature, the lens thickness, the aspheric surface or the back focal length in this embodiment are different from the optical data in the first embodiment. Besides, in this embodiment, the optical axis region **33** of the object-side surface **31** of the third lens element **30** is convex, the periphery region **37** of the image-side surface **32** of the third lens element **30** is concave, the fifth lens element **50** has positive refracting power, the sixth lens element **60** has negative refracting power, the optical axis region **66** of the image-side surface **62** of the sixth lens element **60** is concave, the optical axis region **73** of the object-side surface **71** of the seventh lens element **70** is concave, the optical axis region **76** of the image-side surface **72** of the seventh lens element **70** is convex, the eighth lens element **80** has positive refracting power, the periphery region **84** of the object-side surface **81** of the eighth lens element **80** is concave.

The optical data of the fourth embodiment of the optical imaging lens are shown in FIG. **26** while the aspheric surface data are shown in FIG. **27**. In this embodiment, EFL=7.587 mm; HFOV=40.842 degrees; TTL=9.721 mm; Fno=1.675; ImgH=6.133 mm. In particular: 1. the Fno in this embodiment is smaller than the Fno in the first embodiment; 2. the distortion aberration in this embodiment is better than the distortion aberration in the first embodiment; 3. the thickness ratio of the optical axis region to the periphery region in this embodiment is smaller than that of the optical imaging lens in the first embodiment so it is easier to fabricate to have better yield.

Fifth Embodiment

Please refer to FIG. **14** which illustrates the fifth embodiment of the optical imaging lens **1** of the present invention. Please refer to FIG. **15**A for the longitudinal spherical aberration on the image plane **4** of the fifth embodiment; please refer to FIG. **15**B for the field curvature aberration on the sagittal direction; please refer to FIG. **15**C for the field curvature aberration on the tangential direction, and please refer to FIG. **15**D for the distortion aberration. The components in this embodiment are similar to those in the first embodiment, but the optical data such as the radius of curvature, the lens thickness, the aspheric surface or the back focal length in this embodiment are different from the optical data in the first embodiment. Besides, in this embodiment, the optical axis region **33** of the object-side surface **31** of the third lens element **30** is convex, the periphery region **37** of the image-side surface **32** of the third lens element **30** is concave, the fourth lens element **40** has negative refracting power, the optical axis region **43** of the object-side surface **41** of the fourth lens element **40** concave, the periphery region **84** of the object-side surface **81** of the eighth lens element **80** is concave.

The optical data of the fifth embodiment of the optical imaging lens are shown in FIG. **28** while the aspheric surface data are shown in FIG. **29**. In this embodiment, EFL=7.036 mm; HFOV=40.842 degrees; TTL=9.407 mm; Fno=1.553; ImgH=6.076 mm. In particular: 1. the system length of the optical imaging lens TTL in this embodiment is shorter than the system length of the optical imaging lens TTL in the first embodiment; 2. the Fno in this embodiment is smaller than the Fno in the first embodiment; 3. the distortion aberration in this embodiment is better than the distortion aberration in the first embodiment.

Sixth Embodiment

Please refer to FIG. **16** which illustrates the sixth embodiment of the optical imaging lens **1** of the present invention. Please refer to FIG. **17**A for the longitudinal spherical aberration on the image plane **4** of the sixth embodiment; please refer to FIG. **17**B for the field curvature aberration on the sagittal direction; please refer to FIG. **17**C for the field curvature aberration on the tangential direction, and please refer to FIG. **17**D for the distortion aberration. The components in this embodiment are similar to those in the first embodiment, but the optical data such as the radius of curvature, the lens thickness, the aspheric surface or the back focal length in this embodiment are different from the optical data in the first embodiment. Besides, in this embodiment, the third lens element **30** has positive refracting power, the optical axis region **33** of the object-side surface **31** of the third lens element **30** is convex, the optical axis region **43** of the object-side surface **41** of the fourth lens element **40** is concave, the periphery region **84** of the object-side surface **81** of the eighth lens element **80** is concave.

The optical data of the sixth embodiment of the optical imaging lens are shown in FIG. **30** while the aspheric surface data are shown in FIG. **31**. In this embodiment, EFL=7.031 mm; HFOV=40.842 degrees; TTL=9.118 mm; Fno=1.552; ImgH=5.874 mm. In particular: 1. the system length of the optical imaging lens TTL in this embodiment is shorter than the system length of the optical imaging lens TTL in the first embodiment; 2. the Fno in this embodiment is smaller than the Fno in the first embodiment; 3. the distortion aberration in this embodiment is better than the distortion aberration in the first embodiment.

Seventh Embodiment

Please refer to FIG. **18** which illustrates the seventh embodiment of the optical imaging lens **1** of the present invention. Please refer to FIG. **19**A for the longitudinal spherical aberration on the image plane **4** of the seventh embodiment; please refer to FIG. **19**B for the field curvature aberration on the sagittal direction; please refer to FIG. **19**C for the field curvature aberration on the tangential direction, and please refer to FIG. **19**D for the distortion aberration. The components in this embodiment are similar to those in the first embodiment, but the optical data such as the radius of curvature, the lens thickness, the aspheric surface or the back focal length in this embodiment are different from the optical data in the first embodiment. Besides, in this embodiment, the optical axis region **33** of the object-side surface **31** of the third lens element **30** is convex, the periphery region **37** of the image-side surface **32** of the third lens element **30** is concave, the optical axis region **43** of the object-side surface **41** of the fourth lens element **40** concave, the periphery region **84** of the object-side surface **81** of the eighth lens element **80** is concave.

The optical data of the seventh embodiment of the optical imaging lens are shown in FIG. **32** while the aspheric surface data are shown in FIG. **33**. In this embodiment, EFL=7.030 mm; HFOV=40.845 degrees; TTL=9.216 mm; Fno=1.552; ImgH=5.935 mm. In particular: 1. the system length of the optical imaging lens TTL in this embodiment is shorter than the system length of the optical imaging lens TTL in the first embodiment; 2. the Fno in this embodiment is smaller than the Fno in the first embodiment; 3. the HFOV in this embodiment is larger than the HFOV in the first embodiment; 4. the distortion aberration in this embodiment is better than the distortion aberration in the first embodiment.

Some important ratios in each embodiment are shown in FIG. **34** and in FIG. **35**.

Each embodiment of the present invention provide an optical imaging lens **1** with a small Fno, a larger image height and improved resolution while it is beneficial to maintain the system length, to maintain good imaging quality and to be technically plausible:

1. when the optical imaging lens **1** of the present invention satisfies that the second lens element **20** has negative refracting power, the periphery region **34** of the object-side surface **31** of the third lens element **30** is concave, the seventh lens element **70** has negative refracting power and the periphery region **87** of the image-side surface **82** of the eighth lens element **80** is convex, it is conducive to design a lens with a small f-number and with a large image height.

2. When it is further to limit the optical imaging lens **1** of the present invention with the technical feature combinations of the same efficacy of one of (a)~(c), it is conducive for the increase of the production yield to enhance the producibility:

(a) the third lens element **30** has negative refracting power, the optical axis region **83** of the object-side surface **81** of the eighth lens element **80** is convex and Tavg567/Tstd567≥3.600. The preferable range is

3.600≤Tavg567/Tstd567≤16.000. Wherein, if the third lens element **30** has negative refracting power, it is more conducive to control the turn of rays on the inner field of view (aberration on 0.2 field of view~0.4 field of view). The further limitation that the first lens element **10** has positive refracting power is conducive to reduce TTL;

(b) the optical axis region **53** of the object-side surface **51** of the fifth lens element **50** is convex, the periphery region **54** of the object-side surface **51** of the fifth lens element **50** is concave, the optical axis region **63** of the object-side surface **61** of the sixth lens element **60** is convex and Tavg567/Tstd567≥4.600. The preferable range is 4.600≤Tavg567/Tstd567≤16.000. The further limitation that the first lens element **10** has positive refracting power is conducive to reduce TTL, wherein Tavg567/Tstd567≥4.600 is more conducive to increase the production yield from the fifth lens element **50** to the seventh lens element **70**;

(c) the optical axis region **53** of the object-side surface **51** of the fifth lens element **50** is convex, the periphery region **57** of the image-side surface **52** of the fifth lens element **50** is convex, the optical axis region **63** of the object-side surface **61** of the sixth lens element **60** is convex and Tavg567/Tstd567≥4.600. The preferable range is 4.600≤Tavg567/Tstd567≤16.000, wherein if it goes with the satisfaction that the first lens element **10** has positive refracting power, it is conducive to reduce TTL, wherein Tavg567/Tstd567≥4.600 is more conducive to increase the production yield from the fifth lens element **50** to the seventh lens element **70**.

3. If the optical imaging lens of the present invention satisfies υ3+υ4≥55.000, υ4+υ5≥55.000, υ6+0≤100.000 or υ7+υ8≤100.00, it is conducive to increase the modulation transfer function (MTF) of the optical imaging lens to increase the resolution. The preferable ranges are 55.000≤υ3+υ4≤120.000, 55.000≤υ4+υ5≤120.000, 38.000≤υ6+υ7≤100.000 or 38.000≤υ7+υ8≤100.00. The more preferable range may be 55.000≤υ3+υ4≤95.000, 55.000≤υ4+υ5≤95.000, 60.000≤υ6+υ7≤100.000 or 60.000≤υ7+υ8≤100.00.

4. If the following conditional formulae of the optical imaging lens of the present invention are optionally satisfied, it may keep the thicknesses of and gaps between each lens element in suitable ranges and from that the parameters are too great to shrink the optical imaging lens, or too small to assemble, or the difficulty of the fabrication may be increased while it helps provide a small f-number and a large image height:

1) 7.000≤(EFL+AAG)/BFL, the preferable range is 7.000≤(EFL+AAG)/BFL≤22.000;
2) 4.400≤(TL+ImgH)/D11t32, the preferable range is 4.400≤(TL+ImgH)/D11t32≤7.000;
3) (T1+D12t31)/(G34+T4)≤5.300, the preferable range is 1.000≤(T1+D12t31)/(G34+T4)≤5.300;
4)(D12t31+T3)/(G45+T5)≤3.300, the preferable range is 0.900≤(D12t31+T3)/(G45+T5)≤3.300;
5) 2.800≤Tavg5678/Tstd5678, the preferable range is 2.800≤Tavg5678/Tstd5678≤17.000;
6) D52t71/T7≤2.700, the preferable range is 1.600≤D52t71/T7≤2.700;
7) (D11t32+D52t71)/(T7+G78)≤7.000, the preferable range is 2.000≤(D11t32+D52t71)/(T7+G78)≤7.000;
8) (D52t71+BFL)/(T7+G78)≤3.610, the preferable range is 1.000≤(D52t71+BFL)/(T7+G78)≤3.610;
9) Fno*D11t32/(G78+T8)≤4.000, the preferable range is 2.400≤Fno*D11t32/(G78+T8)≤4.000;
10) Fno*D52t82/D32t52≤5.000, the preferable range is 1.900≤Fno*D52t82/D32t52≤5.000;
11) ALT/D32t52≤5.300, the preferable range is 1.800≤ALT/D32t52≤5.300;
12) 5.800≤(EPD+ImgH)/(Tmax+Tmin), the preferable range is 5.800≤(EPD+ImgH)/(Tmax+Tmin)≤8.000;
13) 5.000≤(TTL+ImgH)/(T1+T3+T8), the preferable range is 5.000≤(TTL+ImgH)/(T1+T3+T8)≤10.000.

In addition, any arbitrary combination of the parameters of the embodiments can be selected to increase the lens limitation so as to facilitate the design of the same structure of the present invention.

In the light of the unpredictability of the optical imaging lens, the present invention suggests the above principles to have a shorter system length of the optical imaging lens, a reduced f-number, enhanced imaging quality or a better fabrication yield to overcome the drawbacks of prior art.

In addition to the above ratios, one or more conditional formulae may be optionally combined to be used in the embodiments of the present invention and the present invention is not limit to this.

In addition to the above ratios, one or more conditional formulae may be optionally combined to be used in the embodiments of the present invention and the present invention is not limit to this. The concave or convex configuration of each lens element or multiple lens elements may be fine-tuned to enhance the performance and/or the resolution. The above limitations may be selectively combined in the embodiments without causing inconsistency.

The contents in the embodiments of the invention include but are not limited to a focal length, a thickness of a lens element, an Abbe number, or other optical parameters. For example, in the embodiments of the invention, an optical parameter A and an optical parameter B are disclosed, wherein the ranges of the optical parameters, comparative relation between the optical parameters, and the range of a conditional expression covered by a plurality of embodiments are specifically explained as follows:

(1) The ranges of the optical parameters are, for example, $\alpha_2 A \le \alpha_1$ or $\beta_2 \le B \le \beta_1$, where α_1 is a maximum value of the optical parameter A among the plurality of embodiments, α_2 is a minimum value of the optical parameter A among the plurality of embodiments, β_1 is a maximum value of the optical parameter B among the plurality of embodiments, and β_2 is a minimum value of the optical parameter B among the plurality of embodiments.

(2) The comparative relation between the optical parameters is that A is greater than B or A is less than B, for example.

(3) The range of a conditional expression covered by a plurality of embodiments is in detail a combination relation or proportional relation obtained by a possible operation of a plurality of optical parameters in each same embodiment. The relation is defined as E, and E is, for example, A+B or A−B or A/B or A*B or $(A*B)^{1/2}$, and E satisfies a conditional expression $E \le_{\gamma_1}$ or E_{γ_2} or $\gamma 2 \le E \le_{\gamma_1}$, where each of γ_1 and γ_2 is a value obtained by an operation of the optical parameter A and the optical parameter B in a same embodiment, γ_1 is a maximum value among the plurality of the embodiments, and γ_2 is a minimum value among the plurality of the embodiments.

The ranges of the aforementioned optical parameters, the aforementioned comparative relations between the optical parameters, and a maximum value, a minimum value, and the numerical range between the maximum value and the minimum value of the aforementioned conditional expressions are all implementable and all belong to the scope disclosed by the invention. The aforementioned description is for exemplary explanation, but the invention is not limited thereto.

The embodiments of the invention are all implementable. In addition, a combination of partial features in a same embodiment can be selected, and the combination of partial features can achieve the unexpected result of the invention with respect to the prior art. The combination of partial features includes but is not limited to the surface shape of a lens element, a refracting power, a conditional expression or the like, or a combination thereof. The description of the embodiments is for explaining the specific embodiments of the principles of the invention, but the invention is not limited thereto. Specifically, the embodiments and the drawings are for exemplifying, but the invention is not limited thereto.

Those skilled in the art will readily observe that numerous modifications and alterations of the device and method may be made while retaining the teachings of the invention. Accordingly, the above disclosure should be construed as limited only by the metes and bounds of the appended claims.

What is claimed is:

1. An optical imaging lens, from an object side to an image side in order along an optical axis comprising: a first lens element, a second lens element, a third lens element, a fourth lens element, a fifth lens element, a sixth lens element, a seventh lens element and an eighth lens element, the first lens element to the eighth lens element each having an object-side surface facing toward the object side and allowing imaging rays to pass through as well as an image-side surface facing toward the image side and allowing the imaging rays to pass through;

the second lens element has negative refracting power;

the third lens element has negative refracting power and a periphery region of the object-side surface of the third lens element is concave;

the seventh lens element has negative refracting power; and

an optical axis region of the object-side surface of the eighth lens element is convex and a periphery region of the image-side surface of the eighth lens element is convex;

wherein lens elements included by the optical imaging lens are only the eight lens elements described above, Tavg567 is an average of three thicknesses from the fifth lens element to the seventh lens element along the optical axis, Tstd567 is a population standard deviation of the three thicknesses from the fifth lens element to the seventh lens element along the optical axis, Fno is a f-number of the optical imaging lens, D11t32 is defined as a distance from the object-side surface of the first lens element to the image-side surface of the third lens element along the optical axis, G78 is an air gap between the seventh lens element and the eighth lens element along the optical axis, T8 is a thickness of the eighth lens element along the optical axis to satisfy the relationship: Tavg567/Tstd567≥3.600 and Fno*D11 t32/(G78+T8) ≤4.000.

2. The optical imaging lens of claim **1**, wherein υ3 is an Abbe number of the third lens element, υ4 is an Abbe number of the fourth lens element, and the optical imaging lens satisfies the relationship: 55.000≤υ3+υ4.

3. The optical imaging lens of claim **1**, wherein υ7 is an Abbe number of the seventh lens element, υ8 is an Abbe number of the eighth lens element, and the optical imaging lens satisfies the relationship: υ7+υ8≤100.000.

4. The optical imaging lens of claim **1**, wherein T1 is a thickness of the first lens element along the optical axis, D12t31 is defined as a distance from the image-side surface of the first lens element to the object-side surface of the third lens element along the optical axis, G34 is an air gap between the third lens element and the fourth lens element along the optical axis, T4 is a thickness of the fourth lens element along the optical axis, and the optical imaging lens satisfies the relationship: (T1+D12t31)/(G34+T4)≤5.300.

5. The optical imaging lens of claim **1**, wherein D52t71 is defined as a distance from the image-side surface of the fifth lens element to the object-side surface of the seventh lens element along the optical axis, T7 is a thickness of the seventh lens element along the optical axis, and the optical imaging lens satisfies the relationship: D52t71/T7≤2.700.

6. The optical imaging lens of claim **1**, wherein EPD is an entrance pupil diameter of the optical imaging lens, ImgH is an image height of the optical imaging lens, Tmax is a maximal lens element thickness from the first lens element to the eighth lens element along the optical axis, Tmin is a minimal lens element thickness from the first lens element to the eighth lens element along the optical axis, and the optical imaging lens satisfies the relationship: 5.800≤(EPD+ImgH)/(Tmax+Tmin).

7. An optical imaging lens, from an object side to an image side in order along an optical axis comprising: a first lens element, a second lens element, a third lens element, a fourth lens element, a fifth lens element, a sixth lens element, a seventh lens element and an eighth lens element, the first lens element to the eighth lens element each having an object-side surface facing toward the object side and allowing imaging rays to pass through as well as an image-side surface facing toward the image side and allowing the imaging rays to pass through;

the second lens element has negative refracting power;

a periphery region of the object-side surface of the third lens element is concave;

an optical axis region of the object-side surface of the fifth lens element is convex and a periphery region of the object-side surface of the fifth lens element is concave;

an optical axis region of the object-side surface of the sixth lens element is convex;

the seventh lens element has negative refracting power; and

a periphery region of the image-side surface of the eighth lens element is convex;

wherein lens elements included by the optical imaging lens are only the eight lens elements described above, Tavg567 is an average of three thicknesses from the fifth lens element to the seventh lens element along the optical axis, Tstd567 is a population standard deviation of the three thicknesses from the fifth lens element to the seventh lens element along the optical axis, D52t71 is defined as a distance from the image-side surface of the fifth lens element to the object-side surface of the seventh lens element along the optical axis, BFL is a distance from the image-side surface of the eighth lens element to an image plane along the optical axis, T7 is a thickness of the seventh lens element along the optical axis, G78 is an air gap between the seventh lens element and the eighth lens element along the optical axis to satisfy the relationship: Tavg567/Tstd567≥**4.600** and (D52t71+BFL)/(T7+G786)≤3.610.

8. The optical imaging lens of claim **7**, wherein υ4 is an Abbe number of the fourth lens element, υ5 is an Abbe number of the fifth lens element, and the optical imaging lens satisfies the relationship: 55.000≤υ4+υ5.

9. The optical imaging lens of claim **7**, wherein EFL is an effective focal length of the optical imaging lens, AAG is a sum of seven air gaps from the first lens element to the eighth lens element along the optical axis, and the optical imaging lens satisfies the relationship: 7.000≤(EFL+AAG)/BFL.

10. The optical imaging lens of claim **8**, wherein D12t31 is defined as a distance from the image-side surface of the first lens element to the object-side surface of the third lens element along the optical axis, T3 is a thickness of the third lens element along the optical axis, G45 is an air gap between the fourth lens element and the fifth lens element along the optical axis, T5 is a thickness of the fifth lens element along the optical axis, and the optical imaging lens satisfies the relationship: (D12t31+T3)/(G45+T5)≤3.300.

11. The optical imaging lens of claim **7**, wherein D11t32 is defined as a distance from the object-side surface of the first lens element to the image-side surface of the third lens element along the optical axis, and the optical imaging lens satisfies the relationship: (D11t32+D52t71)/(T7+G78) ≤7.000.

12. The optical imaging lens of claim **8**, wherein Fno is a f-number of the optical imaging lens, D52t82 is defined as a distance from the image-side surface of the fifth lens element to the image-side surface of the eighth lens element along the optical axis, D32t52 is defined as a distance from the image-side surface of the third lens element to the image-side surface of the fifth lens element along the optical axis, and the optical imaging lens satisfies the relationship: Fno*D52t82/D32t52≤5 0.000.

13. The optical imaging lens of claim **7**, wherein TTL is a distance from the object-side surface of the first lens element to an image plane along the optical axis, ImgH is an image height of the optical imaging lens, T1 is a thickness of the first lens element along the optical axis, T3 is a thickness of the third lens element along the optical axis, T8 is a thickness of the eighth lens element along the optical axis, and the optical imaging lens satisfies the relationship: 5.000≤(TTL+ImgH)/(T1+T3+T8).

14. An optical imaging lens, from an object side to an image side in order along an optical axis comprising: a first lens element, a second lens element, a third lens element, a fourth lens element, a fifth lens element, a sixth lens element, a seventh lens element and an eighth lens element, the first lens element to the eighth lens element each having an object-side surface facing toward the object side and allowing imaging rays to pass through as well as an image-side surface facing toward the image side and allowing the imaging rays to pass through;

the first lens element has positive refracting power;

the second lens element has negative refracting power;

a periphery region of the object-side surface of the third lens element is concave;

an optical axis region of the object-side surface of the fifth lens element is convex and a periphery region of the image-side surface of the fifth lens element is convex;

an optical axis region of the object-side surface of the sixth lens element is convex;

the seventh lens element has negative refracting power; and

a periphery region of the image-side surface of the eighth lens element is convex;

wherein lens elements included by the optical imaging lens are only the eight lens elements described above, Tavg567 is an average of three thicknesses from the fifth lens element to the seventh lens element along the optical axis, Tstd567 is a population standard deviation of the three thicknesses from the fifth lens element to the seventh lens element along the optical axis, D52t71 is defined as a distance from the image-side surface of the fifth lens element to the object-side surface of the seventh lens element along the optical axis, T7 is a thickness of the seventh lens element along the optical axis to satisfy the relationship: Tavg567/Tstd567≥4.600 and D52t71/T7≤2.700.

15. The optical imaging lens of claim **14**, wherein υ**6** is an Abbe number of the sixth lens element, υ7 is an Abbe number of the seventh lens element, and the optical imaging lens satisfies the relationship: υ6+vυ7≤100.000.

16. The optical imaging lens of claim **14**, wherein TL is a distance from the object-side surface of the first lens element to the image-side surface of the eighth lens element along the optical axis, ImgH is an image height of the optical imaging lens, D11t32 is defined as a distance from the object-side surface of the first lens element to the image-side surface of the third lens element along the optical axis, and the optical imaging lens satisfies the relationship: 4.400≤(TL+ImgH)/D11t32.

17. The optical imaging lens of claim **14**, wherein Tavg5678 is an average of four thicknesses from the fifth lens element to the eighth lens element along the optical axis, Tstd5678 is a population standard deviation of the four thicknesses from the fifth lens element to the eighth lens element along the optical axis, and the optical imaging lens satisfies the relationship: 2.800≤Tavg5678/Tstd5678.

18. The optical imaging lens of claim **15**, wherein BFL is a distance from the image-side surface of the eighth lens element to an image plane along the optical axis, G78 is an air gap between the seventh lens element and the eighth lens element along the optical axis, and the optical imaging lens satisfies the relationship: (D52t71+BFL)/(T7+G78)≤3.610.

19. The optical imaging lens of claim **14**, wherein ALT is a sum of the thicknesses of eight lens elements from the first lens element to the eighth lens element along the optical axis, D32t52 is defined as a distance from the image-side surface of the third lens element to the image-side surface of the fifth lens element along the optical axis, and the optical imaging lens satisfies the relationship: ALT/D32t52≤5.300.

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